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ENGINEERING MECHANICS **STATICS**

| SIXTH EDITION |

J. L. MERIAM

L. G. KRAIGE



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5

DISTRIBUTED FORCES

5. Distributed Forces

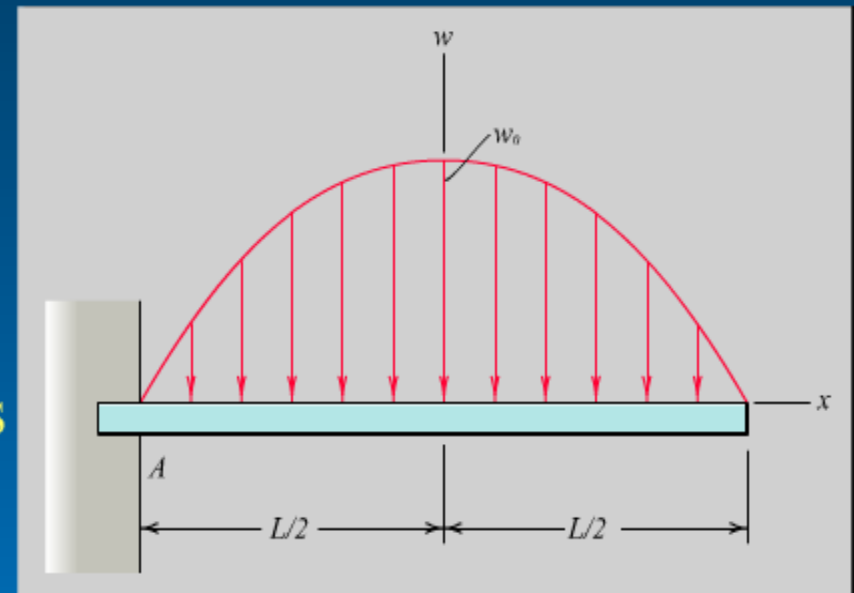
Introduction

Centers of Gravity, Centers of Mass, Centroids

- a. Center of Gravity
- b. Center of Mass
- c. Centroids
- d. Tips for Integration
- e. Composite Techniques

Beams

- a. External Effects
- b. Internal Effects



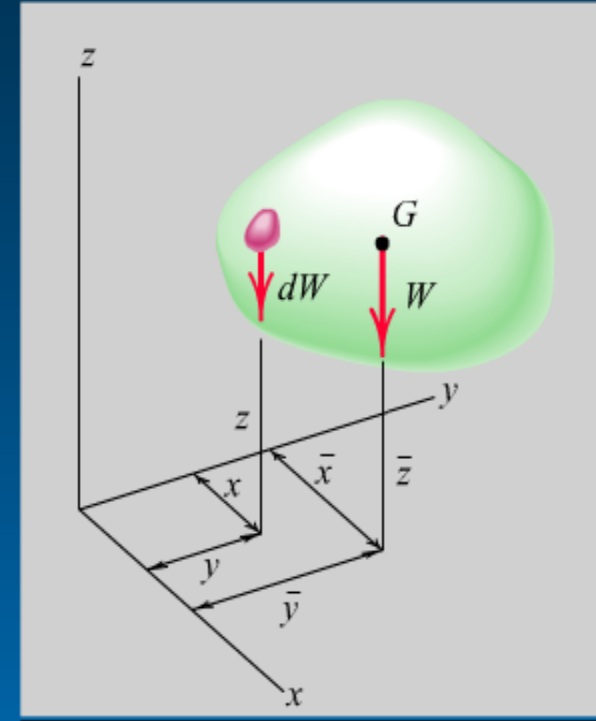
Distributed Forces

Rather than being strictly concentrated, all real forces are spread out, or distributed, to some degree. Forces may be distributed by:

- Line (Example: The weight of a power transmission cable)
- Area (Example: Water pressure against a dam)
- Volume (Example: The weight of the human body)

Center of Gravity

With reference to the figure, we equate the moment of the weight W about the y -axis to the sum of the moments about the y -axis of all the element weights dW , or



$$\bar{x}W = \int x dW \quad \text{or} \quad \bar{x} = \frac{\int x dW}{W}$$

Similarly, with axis relabeling in the case of the z -axis (to make it horizontal), we obtain

$$\bar{y} = \frac{\int y dW}{W} \qquad \bar{z} = \frac{\int z dW}{W}$$

In each case, $W = \int dW$, where all infinite summations are over the entire body. These formulas give the *center of gravity*.

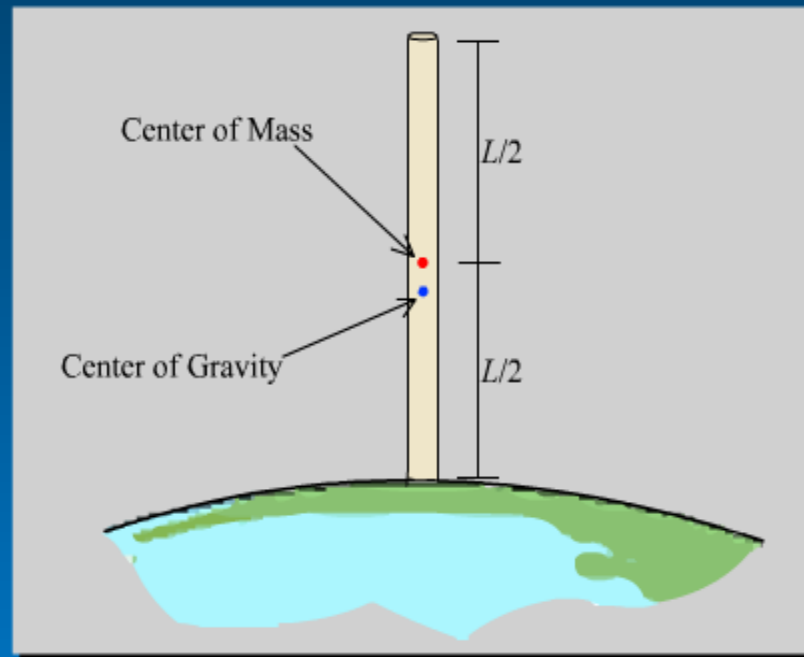
Center of Mass

By substituting $W = mg$ and $dW = g dm$, and letting the acceleration of gravity be constant over the body, we obtain

$$\bar{x} = \frac{\int x dm}{m} \quad \bar{y} = \frac{\int y dm}{m} \quad \bar{z} = \frac{\int z dm}{m}$$

These formulas give the location of the *center of mass*.

Note that the center of gravity and the center of mass, for ordinary sized bodies, are essentially the same. But, due to the variation of g with altitude, the mass center of a mile high uniform metal rod would be at its geometric center, whereas the center of gravity would be a bit lower.



Centroids

In a similar fashion, we can **derive** the centroid locations of volumes, areas, and lines as

Volumes

$$\bar{x} = \frac{\int x dV}{V}$$

$$\bar{y} = \frac{\int y dV}{V}$$

$$\bar{z} = \frac{\int z dV}{V}$$

Areas

$$\bar{x} = \frac{\int x dA}{A}$$

$$\bar{y} = \frac{\int y dA}{A}$$

$$\bar{z} = \frac{\int z dA}{A}$$

Lines

$$\bar{x} = \frac{\int x dL}{L}$$

$$\bar{y} = \frac{\int y dL}{L}$$

$$\bar{z} = \frac{\int z dL}{L}$$

Centroids

In a similar fashion, we can **derive** the centroid locations of volumes, areas, and lines as

Volumes

$$\bar{x} = \frac{\int x dV}{V}$$

$$\bar{y} = \frac{\int y dV}{V}$$

$$\bar{z} = \frac{\int z dV}{V}$$

For example, the first column is derived by beginning with the mass-center formulas and substituting $m = \rho V$ and $dm = \rho dV$, where ρ is the body density, and assuming that ρ does not vary over the body.

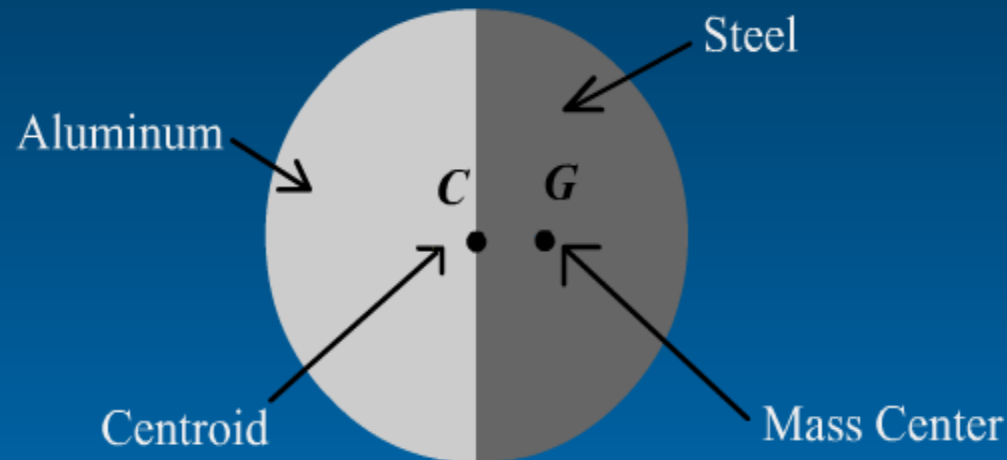


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Centroid is a property of only the geometry of a body, whereas mass center is a function of geometry and mass properties. Consider the disk of half steel and half aluminum:



The centroid of the body is at its geometric center as shown. The mass center is offset to the right.



Tips for Integration

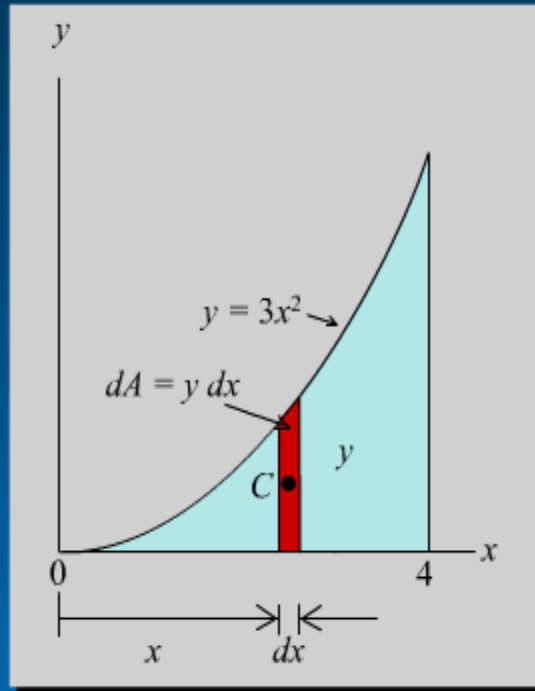
1. Choose the lowest-order element possible. For example, a first-order element allows a single integration, whereas a second-order element requires a double integration.
2. Choose an element which allows one continuous integration to cover the figure.
3. Choose the most convenient coordinates.

4. Realize that the x , y , and z multipliers in the integral expressions of the numerators are coordinates of the *centroid* of the chosen element. We emphasize this with the subscript c :

$$\bar{x} = \frac{\int x_c dA}{\int dA}, \quad \bar{x} = \frac{\int x_c dV}{\int dV}, \quad \text{etc.}$$

In the plane-area example to follow, $y_c = \frac{y}{2}$

Example: Determine the centroid location for the area shown.

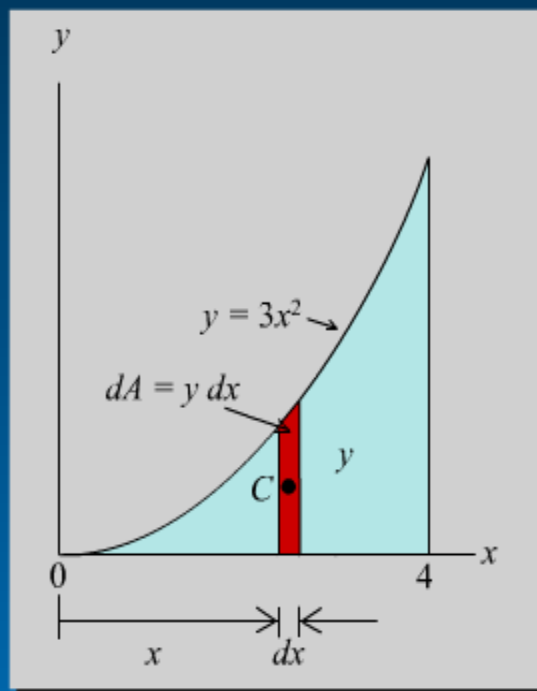


$$\bar{y} = \frac{\int y_c dA}{\int dA} = \frac{\int \frac{y}{2} y dx}{\int y dx}$$

$$= \frac{\frac{1}{2} \int_0^4 (3x^2)^2 dx}{\int_0^4 3x^2 dx} = \frac{\frac{1}{2}(9) \frac{x^5}{5} \Big|_0^4}{3 \frac{x^3}{3} \Big|_0^4}$$

$$= \frac{\frac{9}{10}(4)^5}{(4)^3} = \frac{72}{5}$$

Example: Determine the centroid location for the area shown.



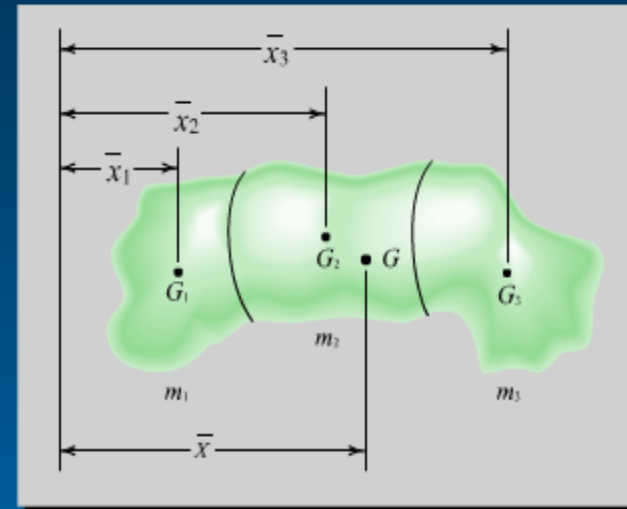
$$\bar{x} = \frac{\int x_c dA}{\int dA} = \frac{\int xy dx}{\int y dx}$$

$$= \frac{\int_0^4 x(3x^2) dx}{\int_0^4 3x^2 dx} = \frac{3 \frac{x^4}{4} \Big|_0^4}{3 \frac{x^3}{3} \Big|_0^4}$$

$$= \frac{3 \cdot 4^4}{4 \cdot 4^3} = 3$$

Composite Techniques

Suppose the body in the figure is divided into three parts as shown. The moment principle gives



$$(m_1 + m_2 + m_3)g\bar{X} = m_1g\bar{x}_1 + m_2g\bar{x}_2 + m_3g\bar{x}_3$$

where $\bar{X}$ is the x -coordinate of the mass center of the entire body.

Generalizing, we have the mass-center relationships

$$\bar{X} = \frac{\sum m\bar{x}}{\sum m} \quad \bar{Y} = \frac{\sum m\bar{y}}{\sum m} \quad \bar{Z} = \frac{\sum m\bar{z}}{\sum m}$$

Similar relationships exist for the centroids of volumes, areas, and lines. For example, for area centroids, we have

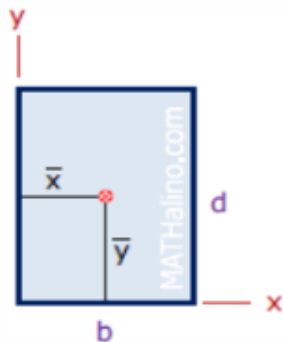
$$\bar{X} = \frac{\sum A\bar{x}}{\sum A} \quad \bar{Y} = \frac{\sum A\bar{y}}{\sum A} \quad \bar{Z} = \frac{\sum A\bar{z}}{\sum A}$$



Centroids of Common Geometric Shapes

Rectangle

Area and Centroid



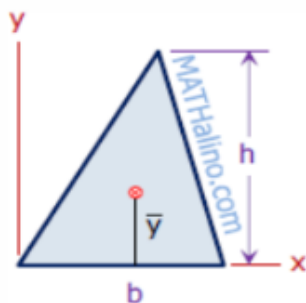
$$A = bd$$

$$\bar{x} = \frac{1}{2}b$$

$$\bar{y} = \frac{1}{2}d$$

Triangle

Area and Centroid

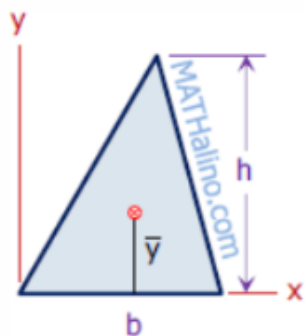


$$A = \frac{1}{2}bh$$

$$\bar{y} = \frac{1}{3}h$$

Triangle

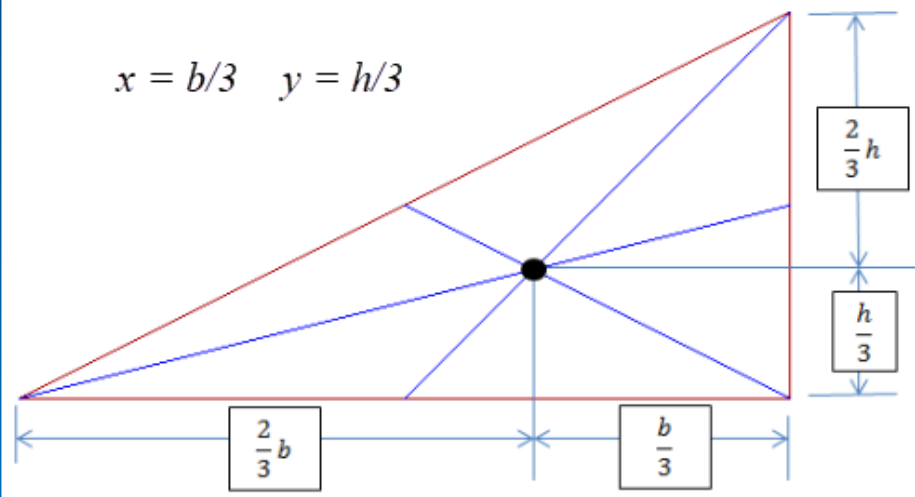
Area and Centroid



$$A = \frac{1}{2}bh$$

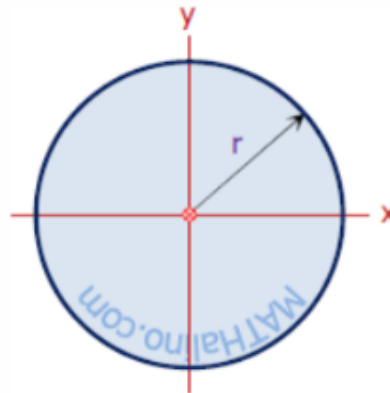
$$\bar{y} = \frac{1}{3}h$$

$$x = \frac{b}{3} \quad y = \frac{h}{3}$$



Circle

Area and Centroid



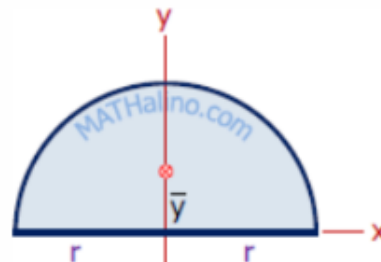
$$A = \pi r^2$$

$$\bar{x} = 0$$

$$\bar{y} = 0$$

Semicircle

Area and Centroid



$$A = \frac{1}{2} \pi r^2$$

$$\bar{x} = 0$$

$$\bar{y} = \frac{4r}{3\pi}$$

Semicircular Arc

Length and Centroid



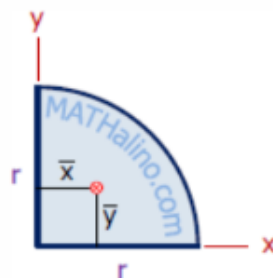
$$L = \pi r$$

$$\bar{x} = \frac{2r}{\pi}$$

$$\bar{y} = 0$$

Quarter Circle

Area and Centroid



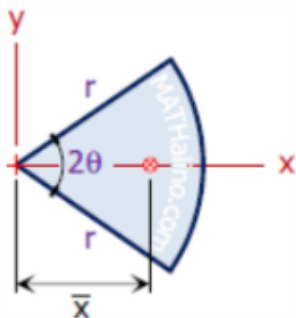
$$A = \frac{1}{4}\pi r^2$$

$$\bar{x} = \frac{4r}{3\pi}$$

$$\bar{y} = \frac{4r}{3\pi}$$

Sector of a Circle

Area and Centroid



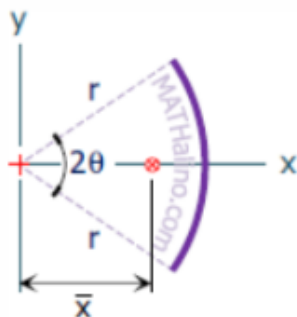
$$A = r^2 \theta_{rad}$$

$$\bar{x} = \frac{2r \sin \theta}{3\theta_{rad}}$$

$$\bar{y} = 0$$

Circular Arc

Length and Centroid



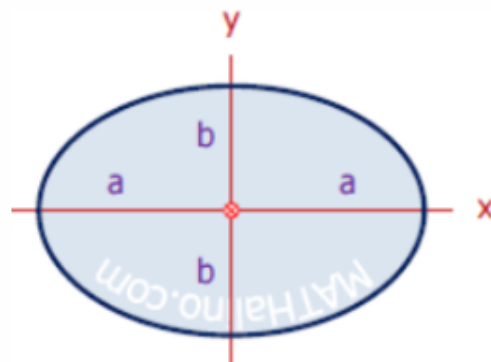
$$L = 2r\theta_{rad}$$

$$\bar{x} = \frac{r \sin \theta}{\theta_{rad}}$$

$$\bar{y} = 0$$

Ellipse

Area and Centroid



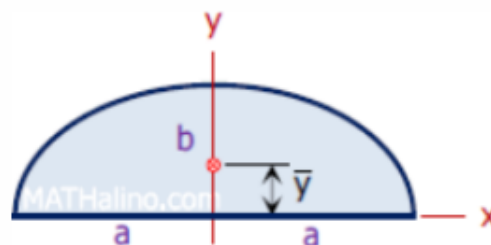
$$A = \pi ab$$

$$\bar{x} = 0$$

$$\bar{y} = 0$$

Half Ellipse

Area and Centroid



$$A = \frac{1}{2} \pi ab$$

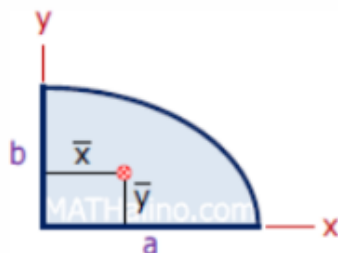
$$\bar{x} = 0$$

$$\bar{y} = \frac{4b}{3\pi}$$

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Quarter Ellipse

Area and Centroid



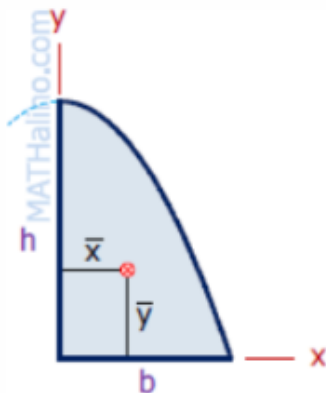
$$A = \frac{1}{4}\pi ab$$

$$\bar{x} = \frac{4a}{3\pi}$$

$$\bar{y} = \frac{4b}{3\pi}$$

Parabolic Segment

Area and Centroid



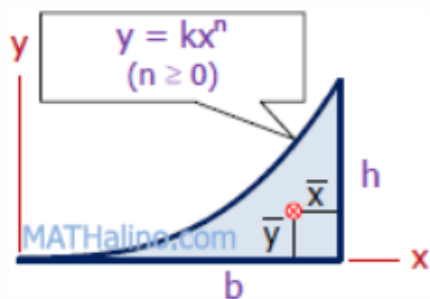
$$A = \frac{2}{3}bh$$

$$\bar{x} = \frac{3}{8}b$$

$$\bar{y} = \frac{2}{5}h$$

Spandrel

Area and Centroid



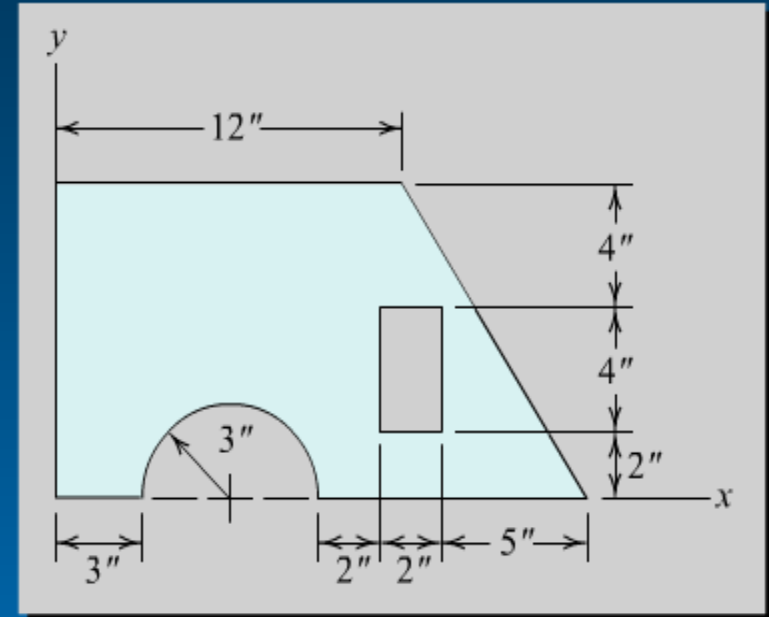
$$A = \frac{1}{n+1}bh$$

$$\bar{x} = \frac{1}{n+2}b$$

$$\bar{y} = \frac{n+1}{4n+2}h$$

Example: Locate the centroid of the shaded area.

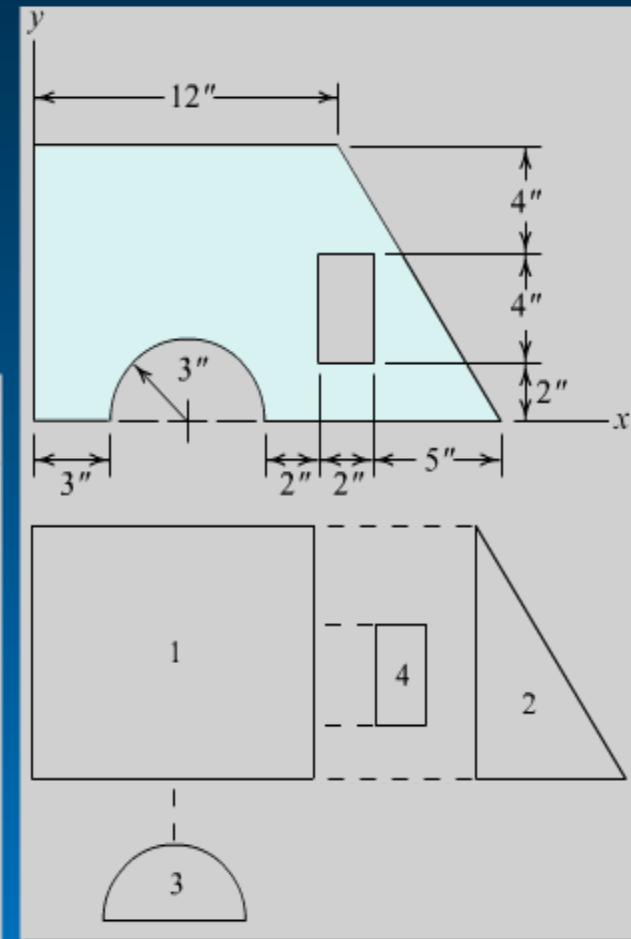
Solution: The composite area is divided into the four elementary shapes shown on the next page. Note that the area of the "holes" (parts 3 and 4) are taken as negative in the following table.



Be sure to reference everything to the same x - y system

Example: Locate the centroid of the shaded area.

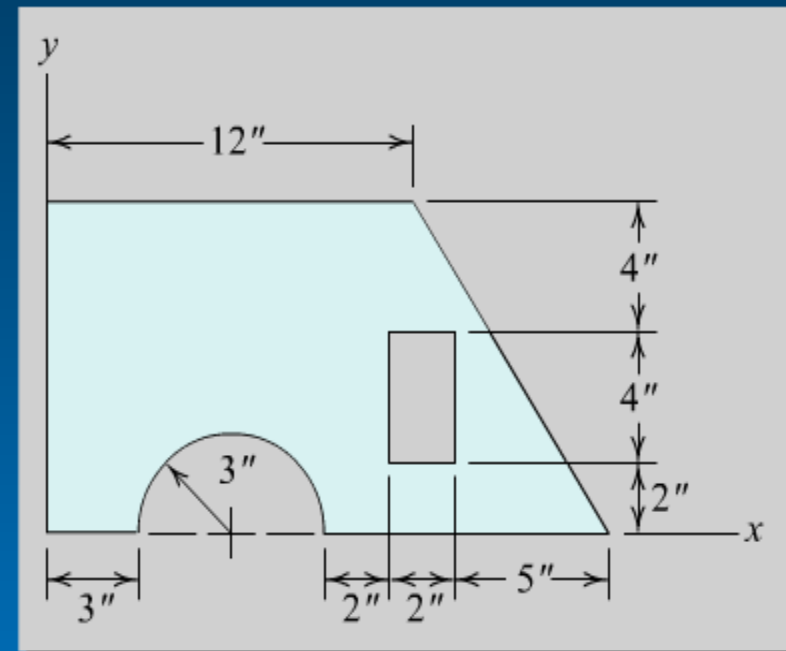
Part	A in. ²	$\bar{x}$ in.	$\bar{y}$ in.	$\bar{x}A$ in. ³	$\bar{y}A$ in. ³
1	120	6	5	720	600
2	30	14	10/3	420	100
3	-14.14	6	1.273	-84.8	-18
4	-8	12	4	-96	-32
Totals	127.9			959	650



Example: Locate the centroid of the shaded area.

Now apply the area centroid equations.

$$\begin{aligned}\bar{X} &= \frac{\Sigma A\bar{x}}{\Sigma A} \\ &= \frac{959}{127.9} = 7.50 \text{ in.} \\ \bar{Y} &= \frac{\Sigma A\bar{y}}{\Sigma A} \\ &= \frac{650}{127.9} = 5.08 \text{ in.}\end{aligned}$$



Sample Problem 5/8

Locate the center of mass of the bracket-and-shaft combination. The vertical face is made from sheet metal which has a mass of 25 kg/m^2 . The material of the horizontal base has a mass of 40 kg/m^2 , and the steel shaft has a density of 7.83 Mg/m^3 .

Solution. The composite body may be considered to be composed of the five elements shown in the lower portion of the illustration. The triangular part will be taken as a negative mass. For the reference axes indicated it is clear by symmetry that the x -coordinate of the center of mass is zero.

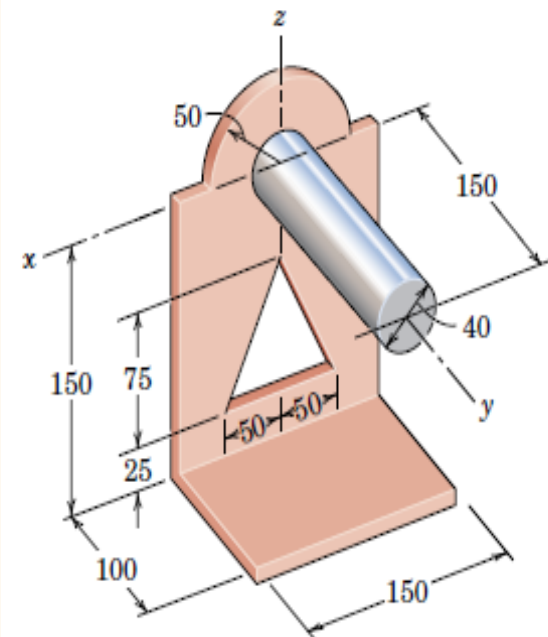
The mass m of each part is easily calculated and should need no further explanation. For Part 1 we have from Sample Problem 5/3

$$\bar{z} = \frac{4r}{3\pi} = \frac{4(50)}{3\pi} = 21.2 \text{ mm}$$

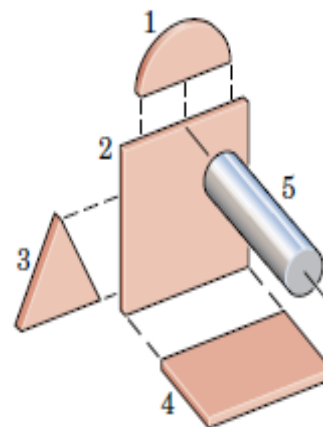
For Part 3 we see from Sample Problem 5/2 that the centroid of the triangular mass is one-third of its altitude above its base. Measurement from the coordinate axes becomes

$$\bar{z} = -[150 - 25 - \frac{1}{3}(75)] = -100 \text{ mm}$$

The y - and z -coordinates to the mass centers of the remaining parts should be evident by inspection. The terms involved in applying Eqs. 5/7 are best handled in the form of a table as follows:



Dimensions in millimeters



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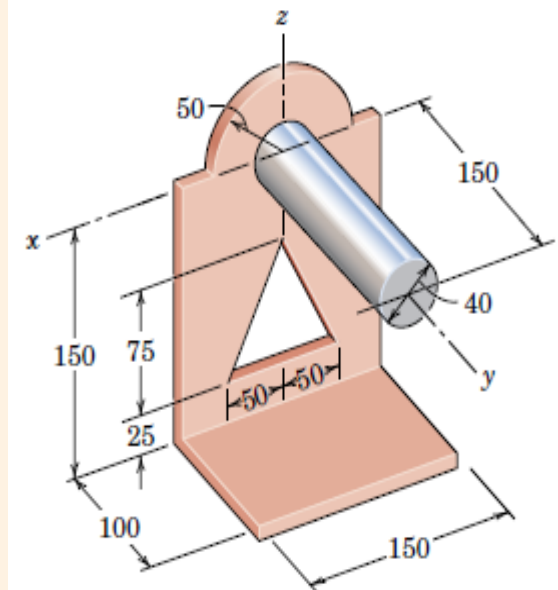
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PART	m kg	$\bar{y}$ mm	$\bar{z}$ mm	$m\bar{y}$ kg·m	$m\bar{z}$ kg·mm
1	0.098	0	21.2	0	2.08
2	0.562	0	-75.0	0	-42.19
3	-0.094	0	-100.0	0	9.38
4	0.600	50.0	-150.0	30.0	-90.00
5	1.476	75.0	0	110.7	0
TOTALS	2.642			140.7	-120.73

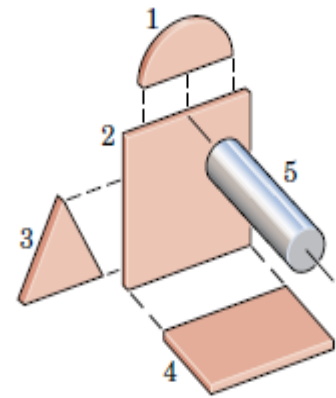
Equations 5/7 are now applied and the results are

$$\left[\bar{Y} = \frac{\Sigma m\bar{y}}{\Sigma m} \right] \quad \bar{Y} = \frac{140.7}{2.642} = 53.3 \text{ mm} \quad \text{Ans.}$$

$$\left[\bar{Z} = \frac{\Sigma m\bar{z}}{\Sigma m} \right] \quad \bar{Z} = \frac{-120.73}{2.642} = -45.7 \text{ mm} \quad \text{Ans.}$$



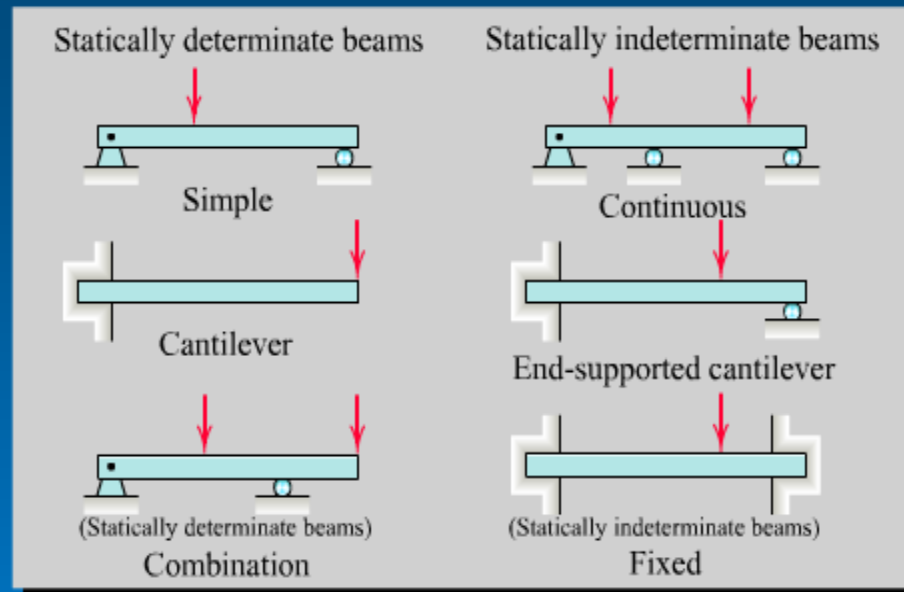
Dimensions in millimeters



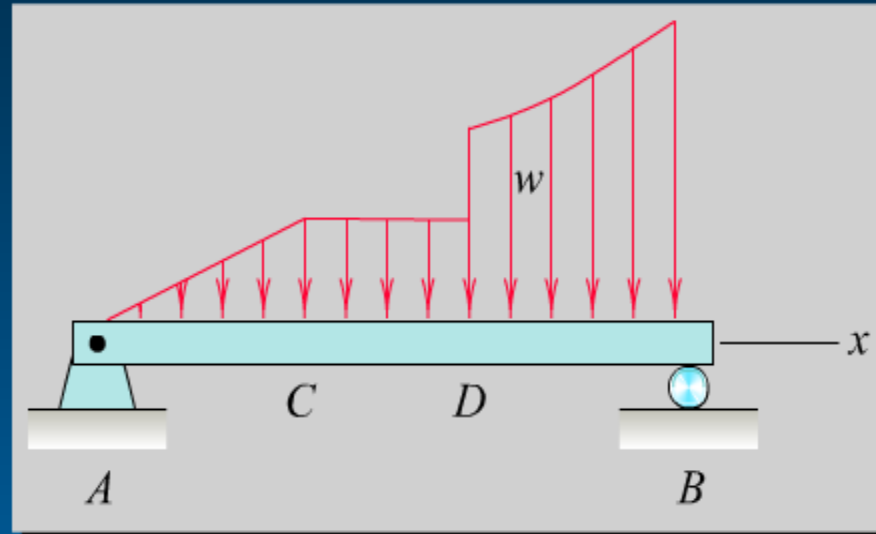
External Effects

Beams are structural members which offer resistance to bending due to applied loads.

Types of beams (with point loads):



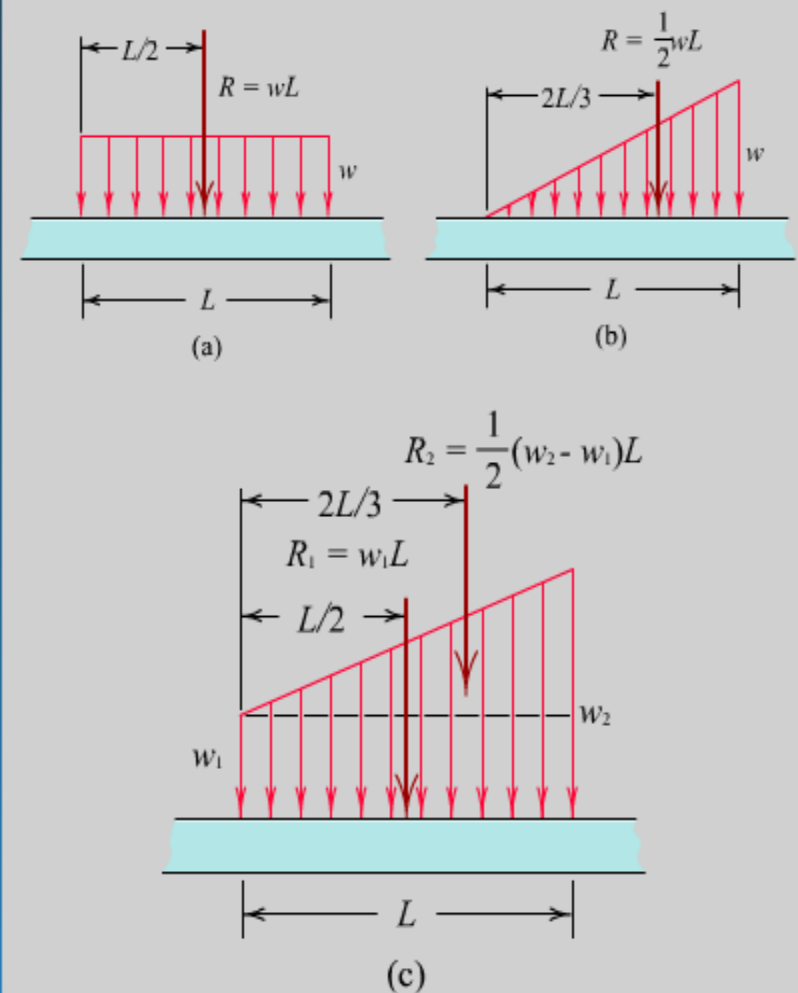
A beam subjected to a distributed load:



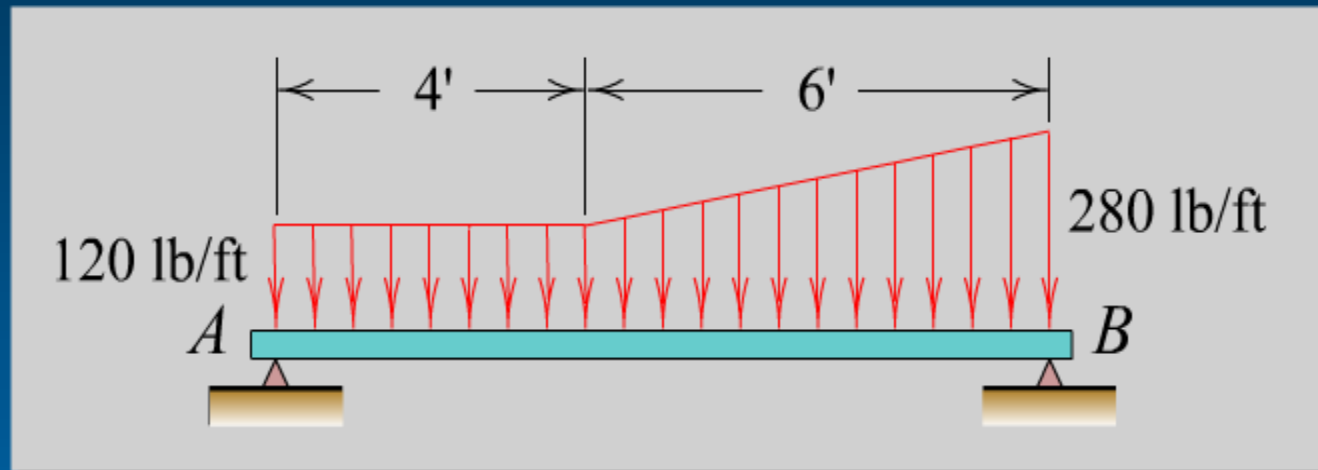
For a beam subjected to a distributed load, we must frequently determine the equivalent point load (both magnitude and location). The magnitude is the *area* under the load-distribution curve, and the location is at the *centroid* of the area.



For the simple load distributions shown, the areas and horizontal centroid locations are easily determined. Note that it is not always necessary to find the centroid of a composite shape like that shown in Figure (c). The individual parts (here a rectangle and a triangle) may be treated separately.



Example: Determine the equivalent concentrated load(s) and external reactions for the simply supported beam which is subjected to the distributed load as shown.



Solution: The area associated with the load distribution is divided into a rectangular and a triangular area. To determine the concentrated load values, compute the areas. The loads are located at the centroids of their respective areas.



Once the concentrated loads are determined, they are placed on the FBD of the beam along with the external reactions at A and B . Now we use principles of equilibrium:

$$\circlearrowleft + \sum M_A = 0:$$

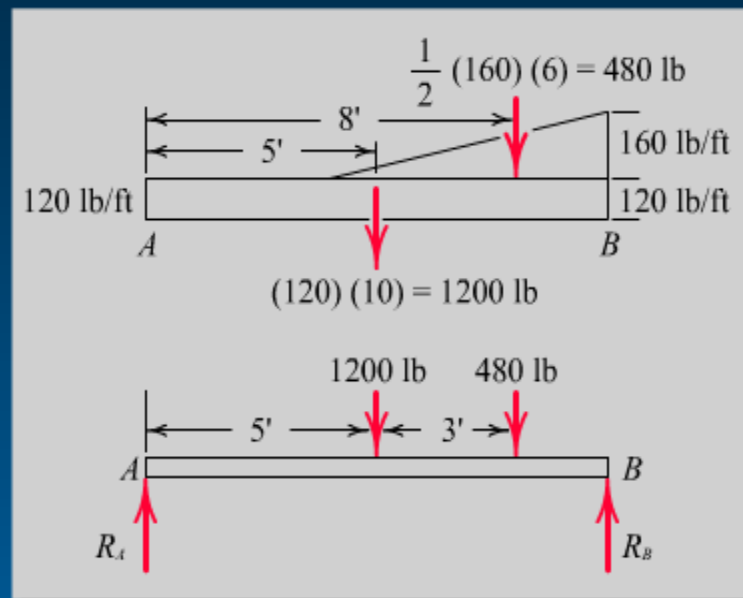
$$1200(5) + 480(8) - R_B(10) = 0$$

$$R_B = 984 \text{ lb}$$

$$\circlearrowleft + \sum M_B = 0:$$

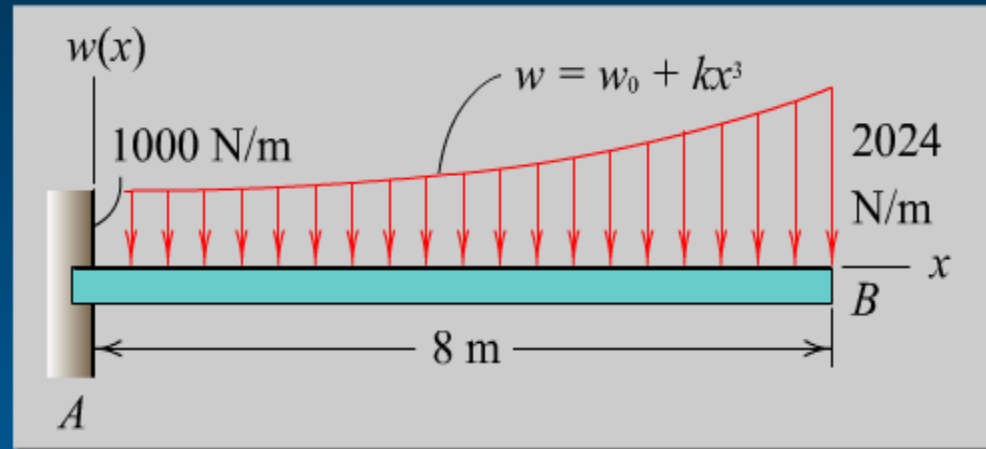
$$R_A(10) - 1200(5) - 480(2) = 0$$

$$R_A = 696 \text{ lb}$$



Example: Determine the reaction at the support A of the loaded cantilever beam.

Solution: The constants in the load distribution are found to be $w_0 = 1000 \text{ N/m}$ and $k = 2 \text{ N/m}^4$. The load R is then



$$R = \int w dx = \int_0^8 (1000 + 2x^3) dx = \left(1000x + \frac{x^4}{2} \right) \Big|_0^8$$
$$= 10048 \text{ N}$$

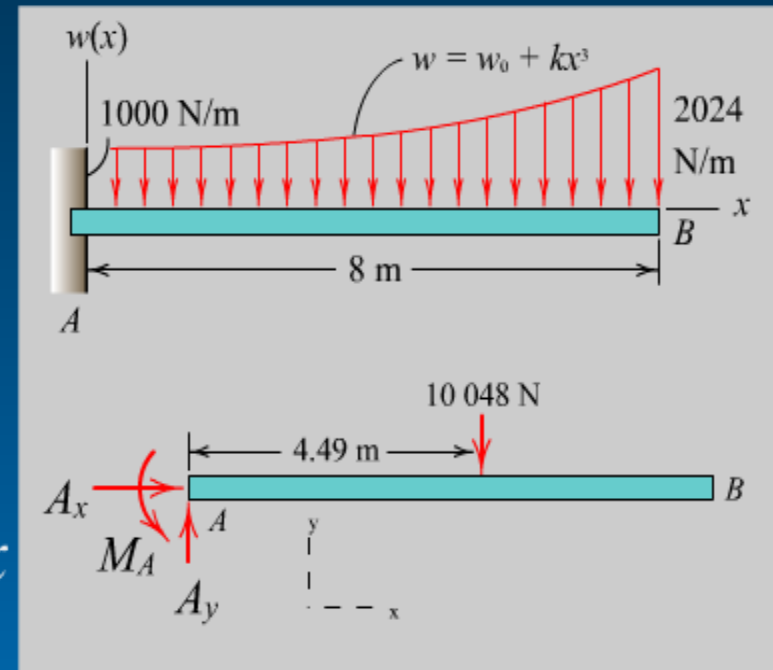
Example: Determine the reaction at the support A of the loaded cantilever beam.

The x -coordinate of the centroid of the area is found by

$$\bar{x} = \frac{\int x w dx}{R}$$

$$= \frac{1}{10048} \int_0^8 x(1000 + 2x^3) dx$$

$$= \frac{1}{10048} \left(500x^2 + \frac{2}{5}x^5 \right) \Big|_0^8 = 4.49 \text{ m}$$



Example: Determine the reaction at the support A of the loaded cantilever beam.

From the free body diagram, we have

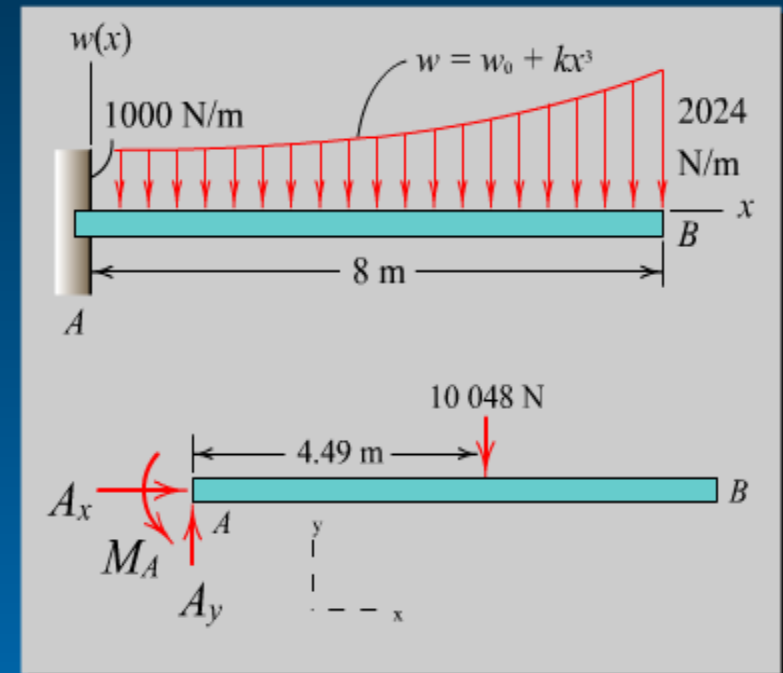
$$\odot^+ \sum M_A = 0$$

$$M_A - (10\,048)(4.49) = 0$$

$$M_A = 45\,100 \text{ N}\cdot\text{m}$$

$$\uparrow^+ \sum F_y = 0$$

$$A_y = 10\,048 \text{ N}$$

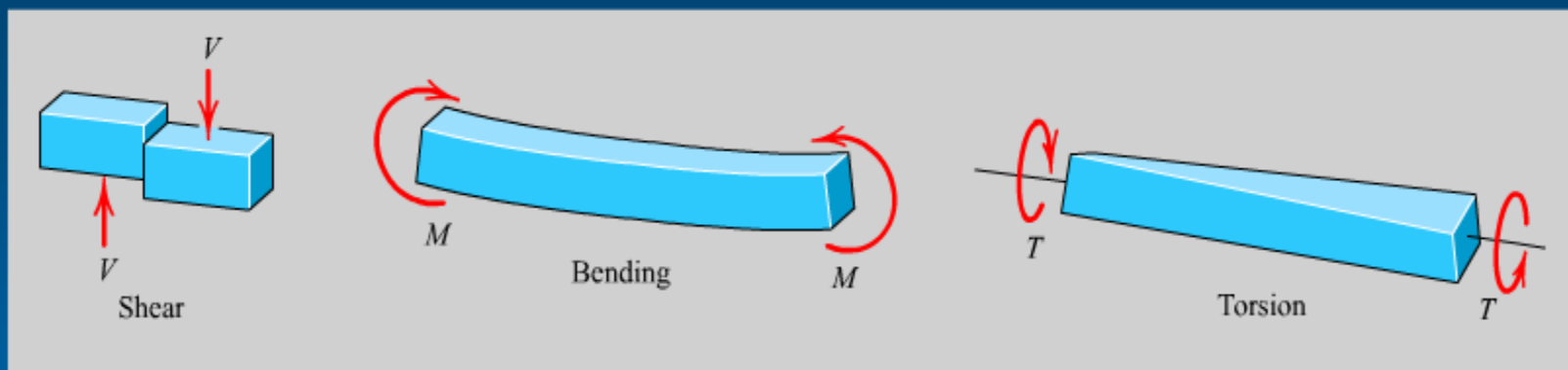


Note that $A_x = 0$ by inspection.



Internal Effects

In addition to supporting tension or compression, a beam can resist *shear*, *bending*, and *torsion*.



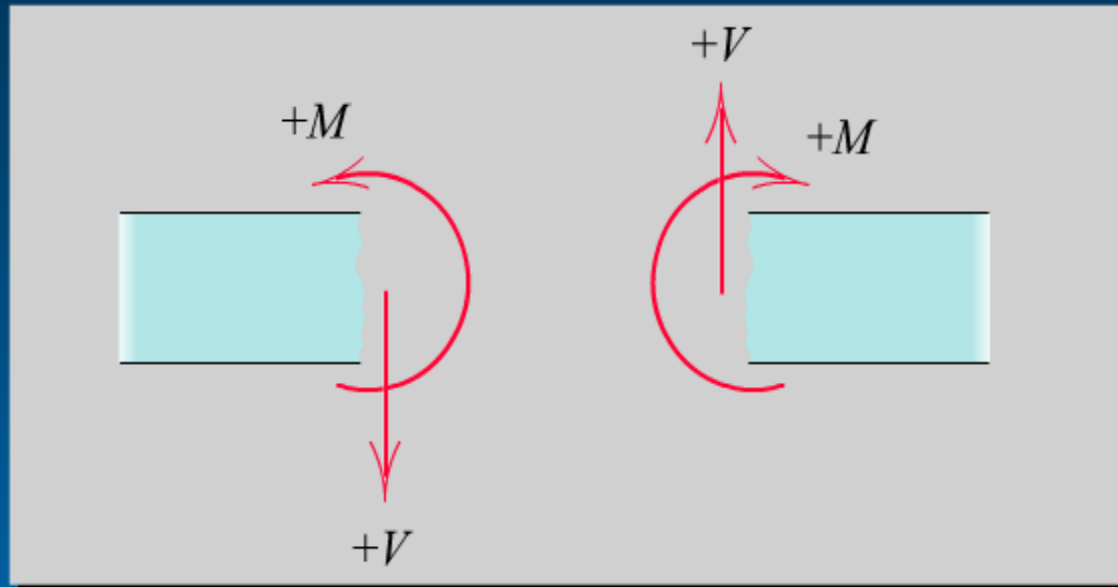
V : Shear Force

M : Bending Moment

T : Torsional Moment

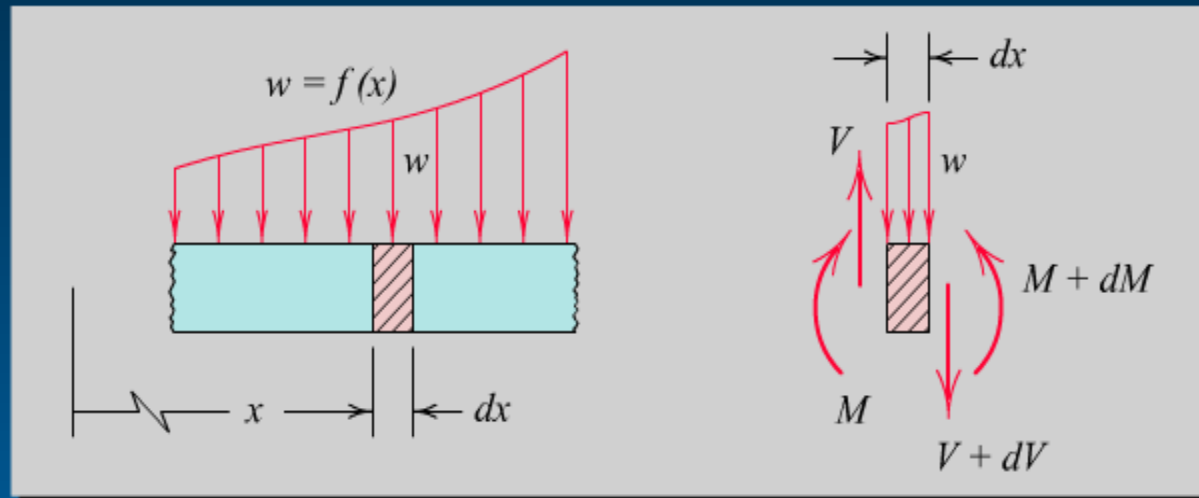


Sign Conventions



The figure shows the convention for positive shear force V and positive bending moment M .

Consider the portion of a loaded beam:

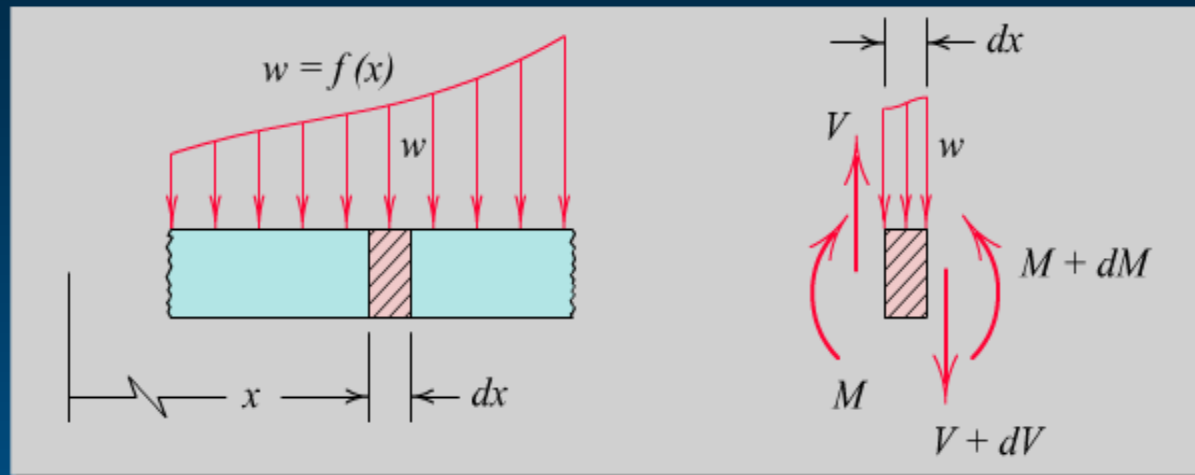


The isolated element has the shear forces and bending moments shown in the assumed positive directions:

Positive shear force: tends to rotate element clockwise

Positive bending moment: tends to bend beam concave upward so as to hold water!





$$+\uparrow \Sigma F = 0 : \quad V - w \, dx - (V + dV) = 0$$

$$\text{or} \quad w = - \frac{dV}{dx}$$

This equation states that the slope of the shear-force diagram must everywhere be equal to the negative of the value of the applied loading. The equation does not hold directly at a concentrated load (where $\left| \frac{dV}{dx} \right| \rightarrow \infty$).



From $w = - \frac{dV}{dx}$

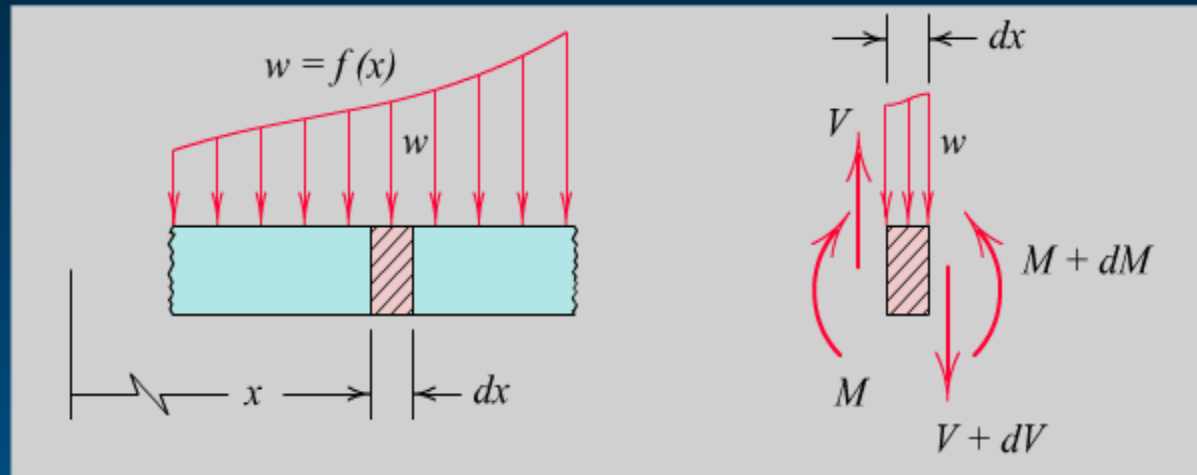
we have $-w \, dx = dV$

$$-\int_{x_0}^x w \, dx = \int_{V_0}^V dV \quad \left(\begin{array}{l} V_0 \text{ is shear force at } x_0 \\ V \text{ is shear force at } x \end{array} \right)$$

or

$$V = V_0 - \int_{x_0}^x w \, dx$$

So $V = V_0 +$ (the negative of the area under the loading curve from x_0 to x)



$$+\circlearrowleft \sum M_{\text{left side}} = 0 : M + w dx \left(\frac{dx}{2} \right) + (V + dV)dx - (M + dM) = 0$$

Neglecting products of differentials, we obtain

$$V = \frac{dM}{dx}$$

This equation states that the shear force is everywhere equal to the slope of the bending-moment curve. The equation does not hold directly at a concentrated couple (where $\left| \frac{dM}{dx} \right| \rightarrow \infty$).



From $V = \frac{dM}{dx}$

we have $Vdx = dM$

$$\int_{x_0}^x Vdx = \int_{M_0}^M dM \quad \left(\begin{array}{l} M_0 \text{ is bending moment at } x_0 \\ M \text{ is bending moment at } x \end{array} \right)$$

or

$$M = M_0 + \int_{x_0}^x Vdx$$

So $M = M_0 + (\text{area under shear-force curve from } x_0 \text{ to } x)$



7/10



Review :

$$w = - \frac{dV}{dx}$$

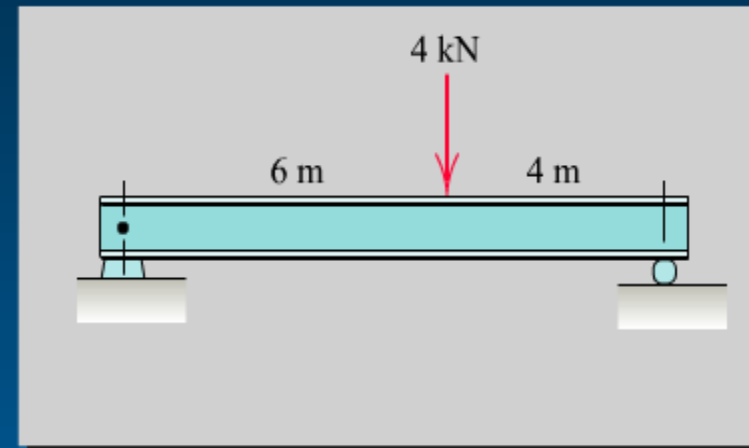
$$V = \frac{dM}{dx}$$

Combining the two equations :

$$\frac{d^2M}{dx^2} = - w$$

Example 1: Determine the shear and moment distribution produced in the simple beam by the 4-kN concentrated load.

Step 1: Find support reactions



Step 2: Find V and M left of force

Step 3: Find V and M right of force

Step 4: Shear and bending moment diagrams

Click on yellow steps to show solutions



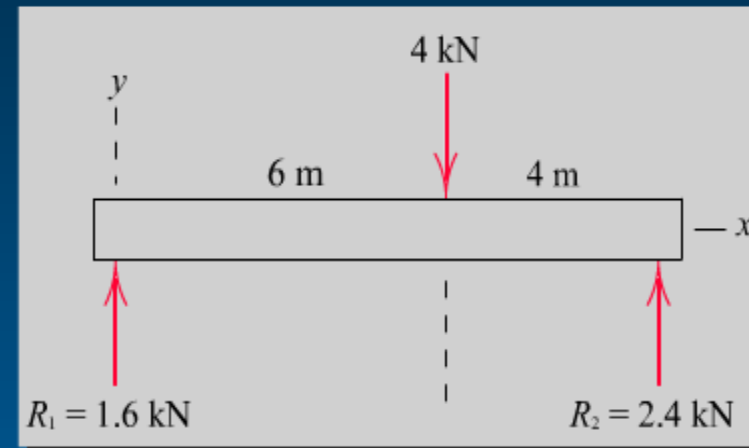
9/10



Example 1: Determine the shear and moment distribution produced in the simple beam by the 4-kN concentrated load.

Step 1: Find support reactions

Using the free-body diagram of the entire beam, we find the support reactions.



$$\Sigma M_{\text{Left}}: 4(6) - 10R_2 = 0 \quad R_2 = 2.4 \text{ kN}$$

$$\Sigma F: R_1 - 4 + R_2 = 0 \quad R_1 = 1.6 \text{ kN}$$

Step 2: Find V and M left of force

Step 3: Find V and M right of force

Step 4: Shear and bending moment diagrams

Click on yellow steps to show solutions



Step 1: Find support reactions

$$R_1 = 1.6 \text{ kN} \quad R_2 = 2.4 \text{ kN}$$

Step 2: Find V and M left of force

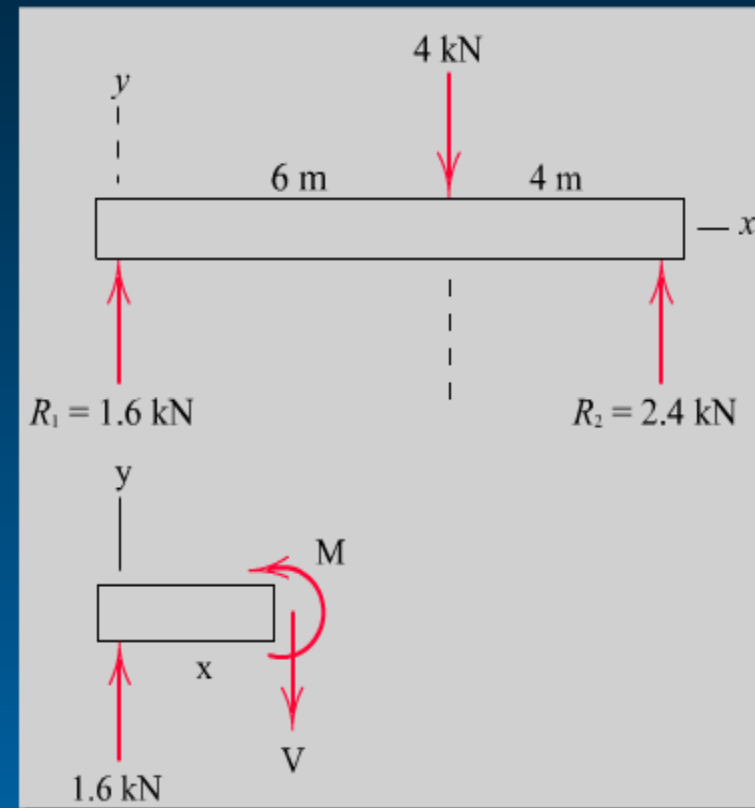
Isolate a section of the beam, length x . V and M are in the positive directions.

$$\Sigma F_y: 1.6 - V = 0$$

$$V = 1.6 \text{ kN}$$

$$\Sigma M_{\text{Right}}: M - 1.6x = 0$$

$$M = 1.6x \text{ kN}\cdot\text{m}$$



Step 3: Find V and M right of force

Step 4: Shear and bending moment diagrams

Click on yellow steps to show solutions



9/10



Step 1: Find support reactions

$$R_1 = 1.6 \text{ kN} \quad R_2 = 2.4 \text{ kN}$$

Step 2: Find V and M left of force

$$0 \text{ m} < x < 6 \text{ m}$$

$$V = 1.6 \text{ kN} \quad M = 1.6x \text{ kN}\cdot\text{m}$$

Step 3: Find V and M right of force

Isolate a section of the beam to the right of the force. V and M are in the positive directions.

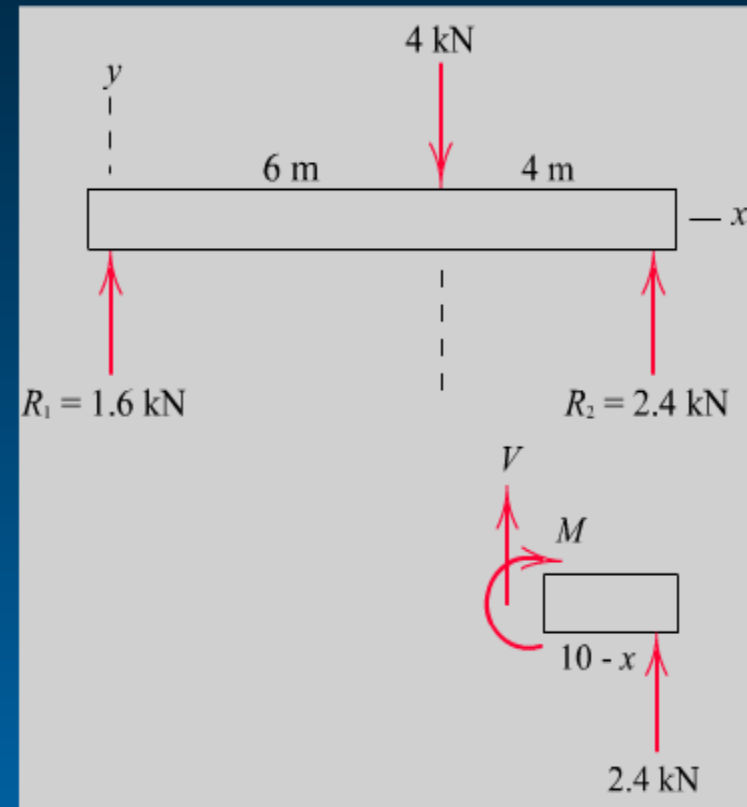
$$\Sigma V: V + 2.4 = 0$$

$$V = -2.4 \text{ kN}$$

$$\Sigma M_{\text{Left}}: -(2.4)(10-x) + M = 0 \quad M = 2.4(10-x) \text{ kN}\cdot\text{m}$$

Step 4: Shear and bending moment diagrams

Click on yellow steps to show solutions



Step 1: Find support reactions

$$R_1 = 1.6 \text{ kN} \quad R_2 = 2.4 \text{ kN}$$

Step 2: Find V and M left of force

$$0 \text{ m} < x < 6 \text{ m}$$

$$V = 1.6 \text{ kN} \quad M = 1.6x \text{ kN}\cdot\text{m}$$

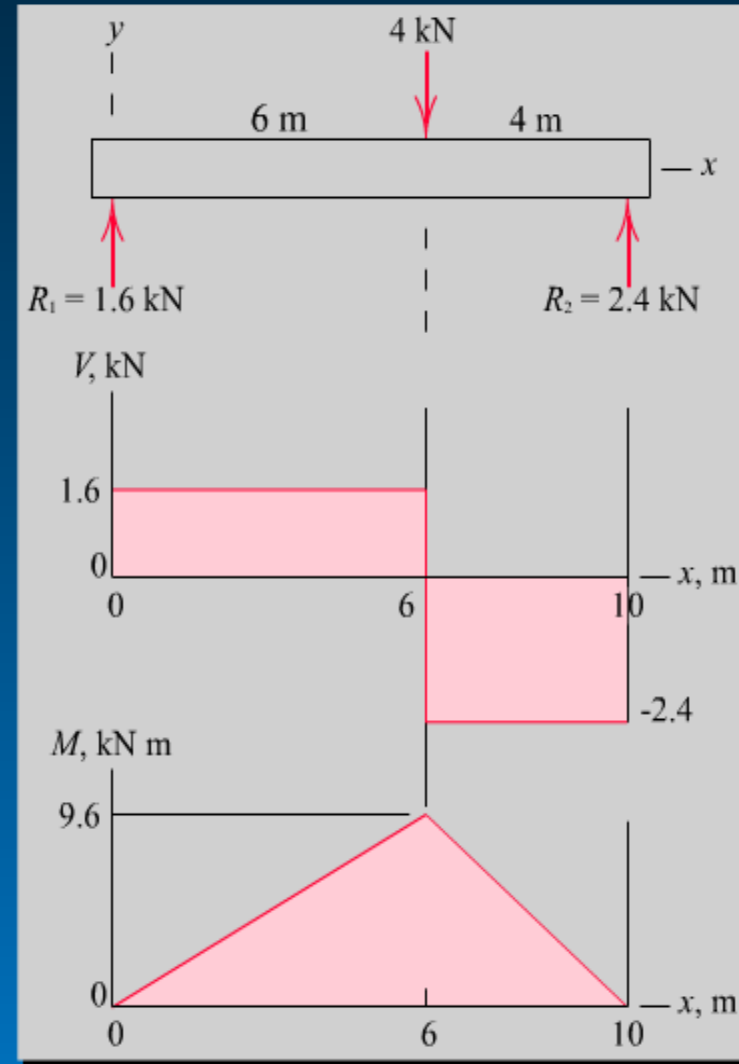
Step 3: Find V and M right of force

$$6 \text{ m} < x < 10 \text{ m}$$

$$V = -2.4 \text{ kN} \quad M = 2.4(10-x) \text{ kN}\cdot\text{m}$$

Step 4: Shear and bending moment diagrams

Comments



Click on yellow steps to show solutions



9/10

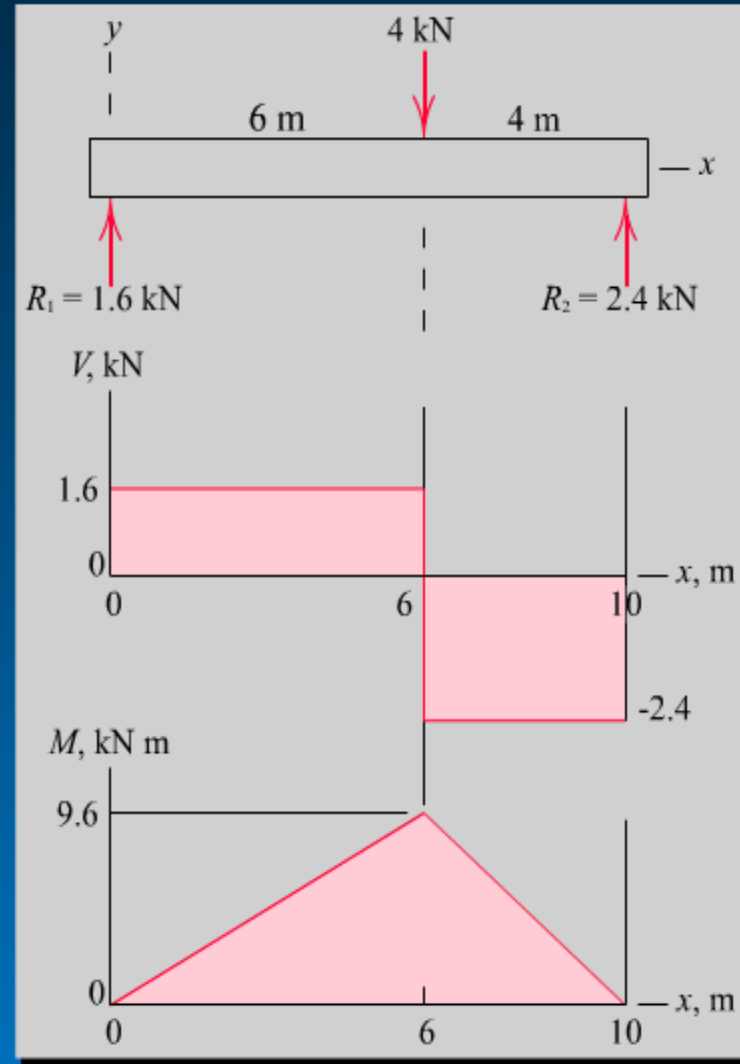


Shear and bending moment diagrams

Comments:

The maximum bending moment occurs where the shear changes direction.

Starting at $x = 0$ and moving right, we see that the moment M is the accumulated area under the shear diagram.



Example 2: Draw the shear-force and bending-moment diagrams for the loaded beam. Determine the maximum moment M and its location x from the left end.

Step 1: Find support reactions

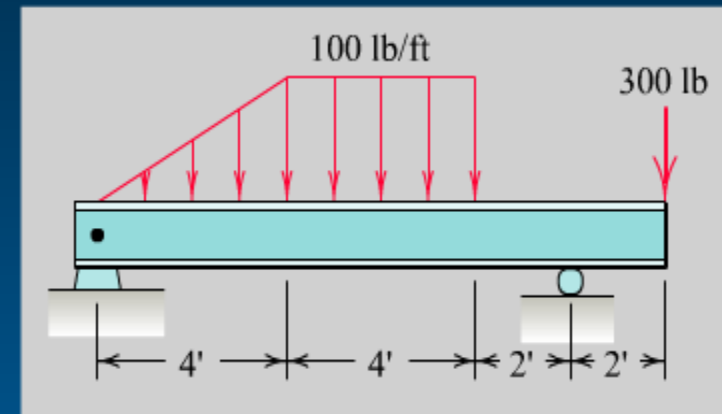
Step 2: Find V and M for $0 < x < 4$ ft

Step 3: Find V and M for $4 < x < 8$ ft

Step 4: Find V and M for $8 < x < 10$ ft

Step 5: Find V and M for $10 < x < 12$ ft

Step 6: Find $M_{\max}$ and its location



Click on yellow steps to show solutions



10/10



Step 1: Find support reactions

The support reactions are obtained using the resultants of the distributed forces on the FBD.

$$\Sigma M_{\text{Left}}: 200\left(\frac{8}{3}\right) + 400(6) - R_2(10) + 300(12) = 0 \quad R_2 = 653 \text{ lb}$$

$$\Sigma F_y: R_1 - 200 - 400 - 300 + R_2 = 0$$

$$R_1 = 247 \text{ lb}$$

Step 2: Find V and M for $0 < x < 4$ ft

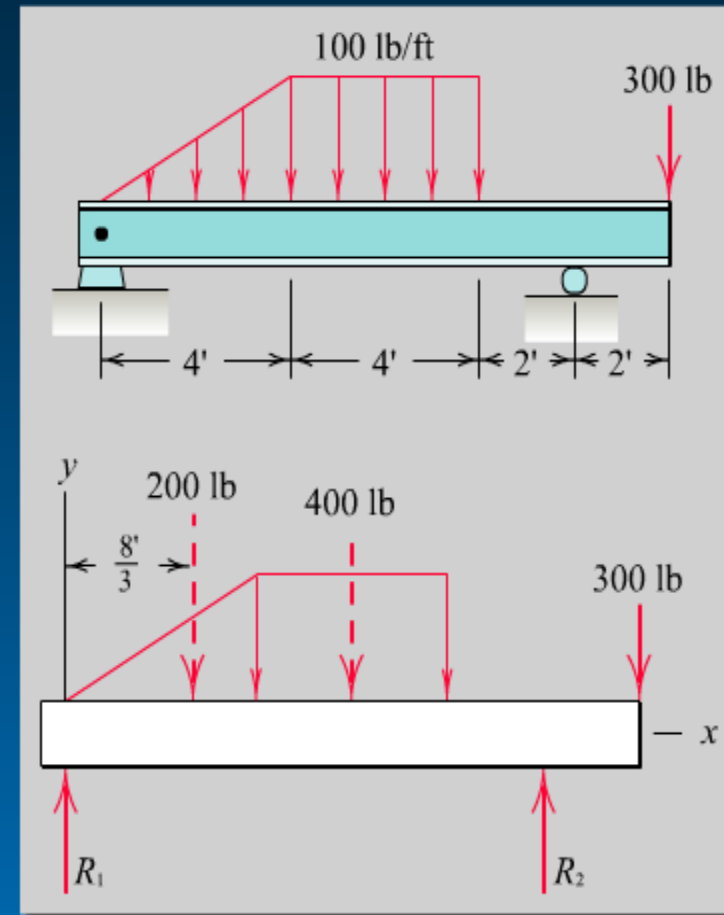
Step 3: Find V and M for $4 < x < 8$ ft

Step 4: Find V and M for $8 < x < 10$ ft

Step 5: Find V and M for $10 < x < 12$ ft

Step 6: Find M_{max} and its location

Click on yellow steps to show solutions



Step 1: Find support reactions

Step 2: Find V and M for $0 < x < 4$ ft

To find the internal reactions we use the FBD of the section.

$$\Sigma F_y: 12.5x^2 + V - 247 = 0$$

$$V = 247 - 12.5x^2 \text{ lb}$$

$$\Sigma M_{\text{Right}}: M + (12.5x^2) \frac{x}{3} - 247x = 0$$

$$M = 247x - 4.17x^3 \text{ lb-ft}$$

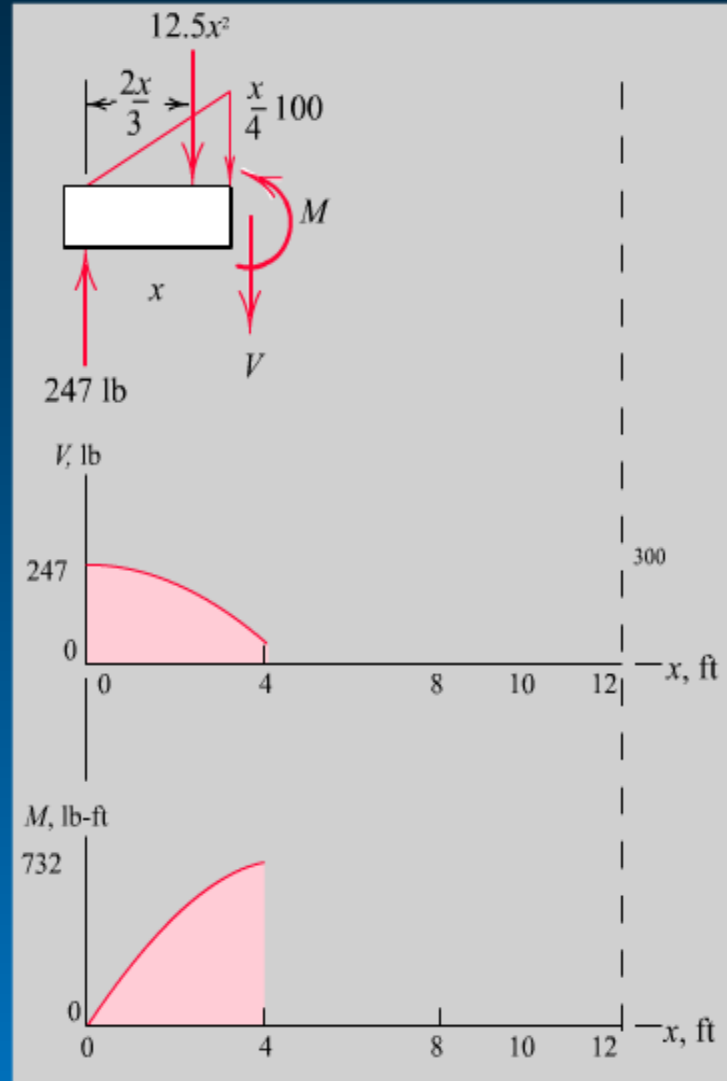
Step 3: Find V and M for $4 < x < 8$ ft

Step 4: Find V and M for $8 < x < 10$ ft

Step 5: Find V and M for $10 < x < 12$ ft

Step 6: Find M_{max} and its location

Click on yellow steps to show solutions



Step 1: Find support reactions

Step 2: Find V and M for $0 < x < 4$ ft

Step 3: Find V and M for $4 < x < 8$ ft

From the FBD, we have:

$$\Sigma F_y: V + 100(x - 4) + 200 - 247 = 0$$

$$V = 447 - 100x \text{ lb}$$

$$\Sigma M_{\text{Right}}: M + 100(x - 4) \frac{x - 4}{2} + 200\left[x - \frac{2}{3}(4)\right] = 0$$

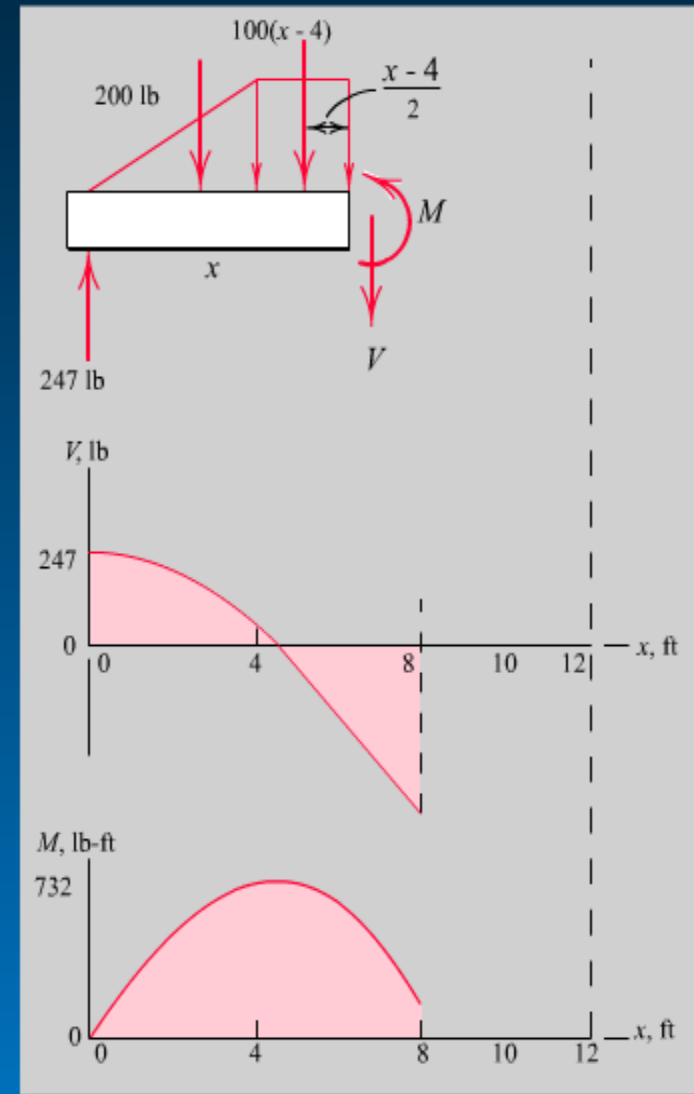
$$M = -267 + 447x - 50x^2 \text{ lb-ft}$$

Step 4: Find V and M for $8 < x < 10$ ft

Step 5: Find V and M for $10 < x < 12$ ft

Step 6: Find M_{max} and its location

Click on yellow steps to show solutions



Step 1: Find support reactions

Step 2: Find V and M for $0 < x < 4$ ft

Step 3: Find V and M for $4 < x < 8$ ft

Step 4: Find V and M for $8 < x < 10$ ft

From the FBD of the right side of the beam:

$$\Sigma F_y: V + 653 - 300 = 0$$

$$V = -353 \text{ lb}$$

$$\Sigma M_{\text{Left}}: M - 653(10 - x) +$$

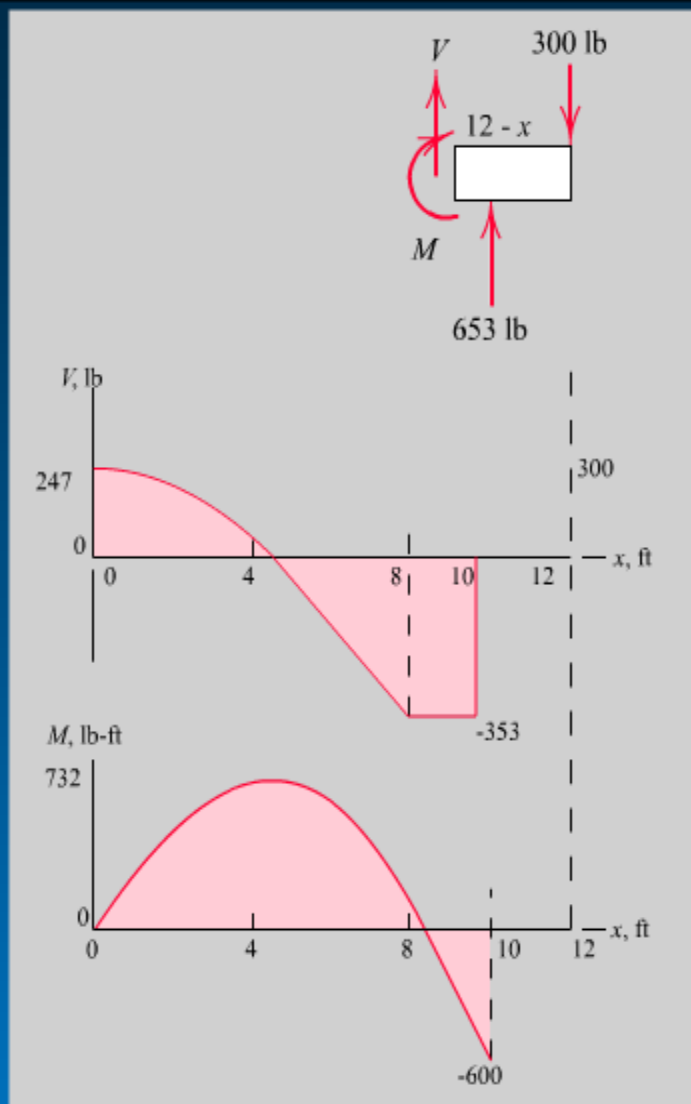
$$300(12 - x) = 0$$

$$M = 2930 - 353x \text{ lb-ft}$$

Step 5: Find V and M for $10 < x < 12$ ft

Step 6: Find M_{max} and its location

Click on yellow steps to show solutions

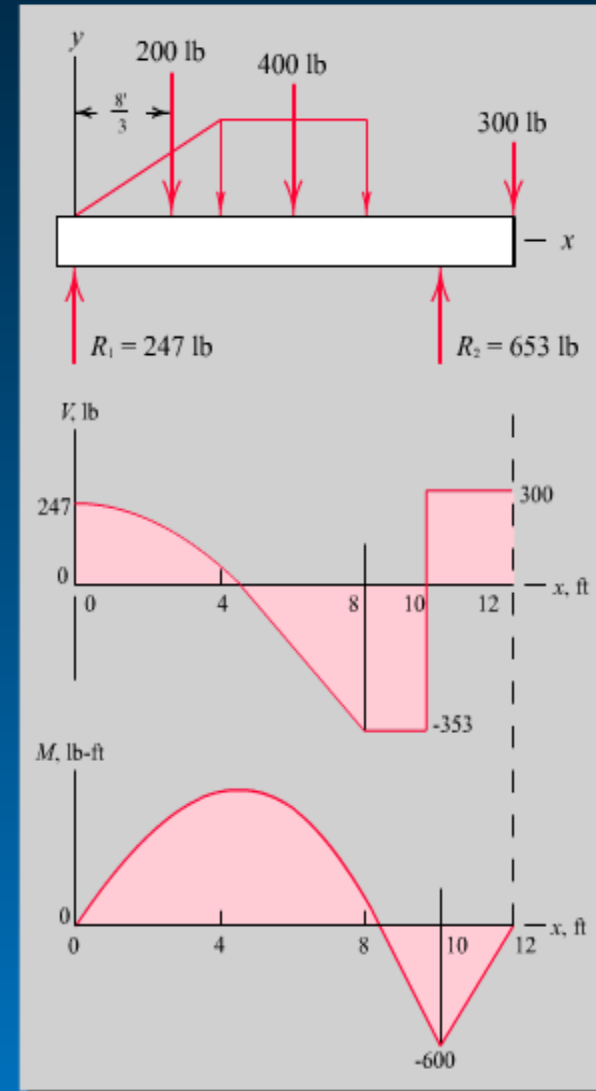


- Step 1: Find support reactions
Step 2: Find V and M for $0 < x < 4$ ft
Step 3: Find V and M for $4 < x < 8$ ft
Step 4: Find V and M for $8 < x < 10$ ft
Step 5: Find V and M for $10 < x < 12$ ft

The last section may be analyzed by inspection. The shear is constant at +300 lb. The moment follows a straight-line relation beginning with zero at the right end of the beam.

Step 6: Find $M_{\max}$ and its location

Click on yellow steps to show solutions



Step 1: Find support reactions

Step 2: Find V and M for $0 < x < 4$ ft

Step 3: Find V and M for $4 < x < 8$ ft

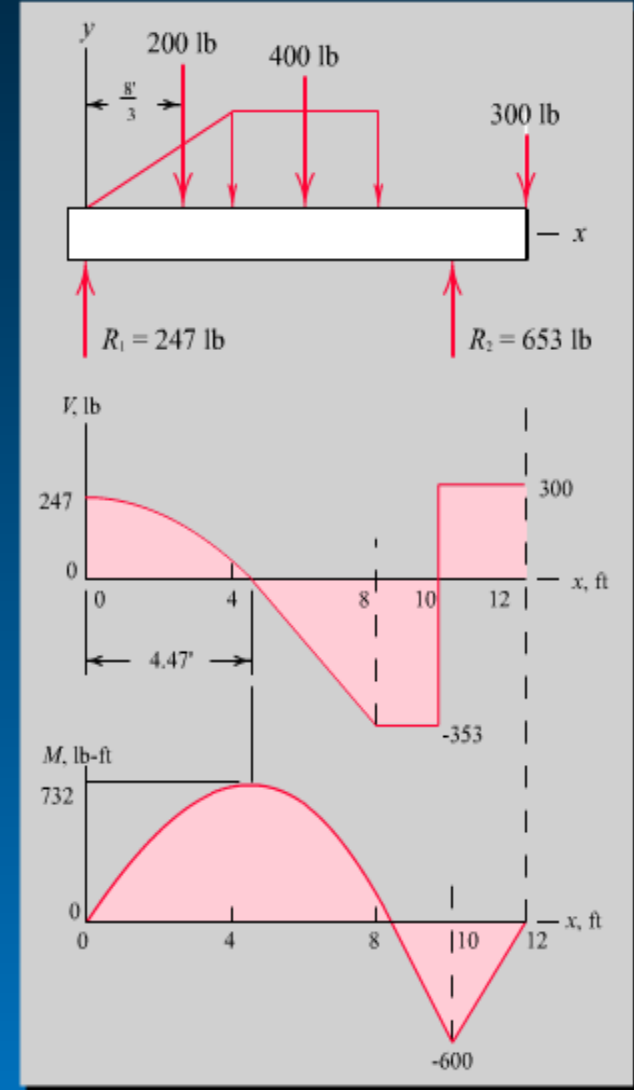
Step 4: Find V and M for $8 < x < 10$ ft

Step 5: Find V and M for $10 < x < 12$ ft

Step 6: Find $M_{\max}$ and its location

The maximum moment occurs at $x = 4.47$ ft, where the shear curve crosses the x -axis. The magnitude of M is obtained by substituting 4.47 into the expression for M for the second interval.

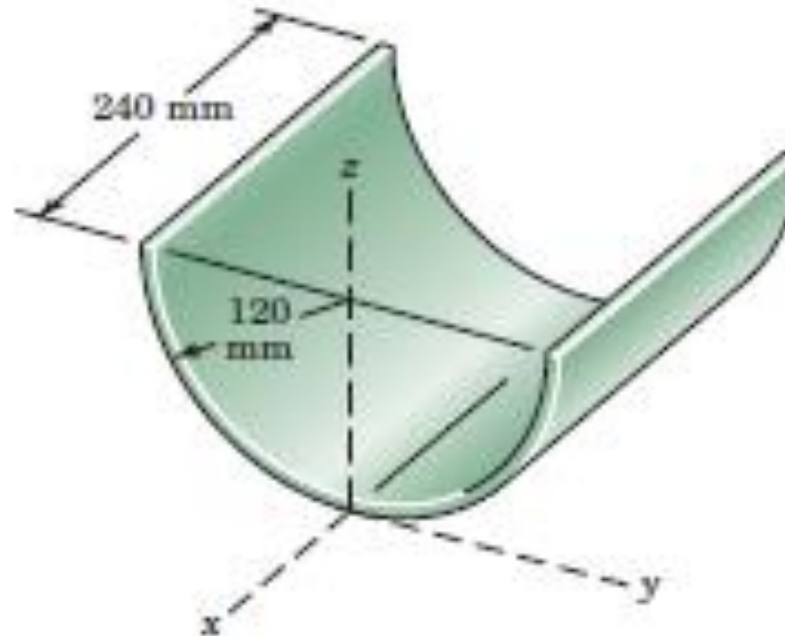
$$M_{\max} = 732 \text{ lb-ft}$$



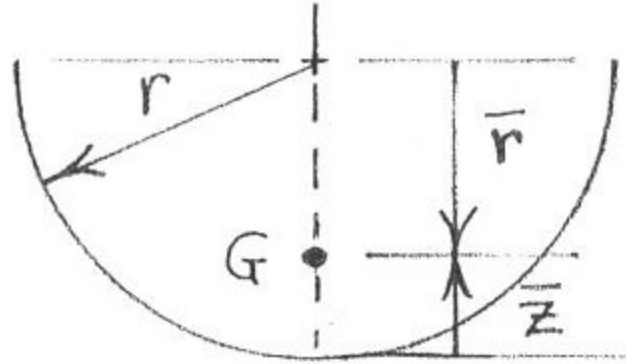
CENTROIDS OF LINES, AREAS, AND VOLUMES

5/3 Specify the x - and z -coordinates of the center of mass of the semicylindrical shell.

Ans. $\bar{x} = -120$ mm, $\bar{z} = 43.6$ mm



5/3



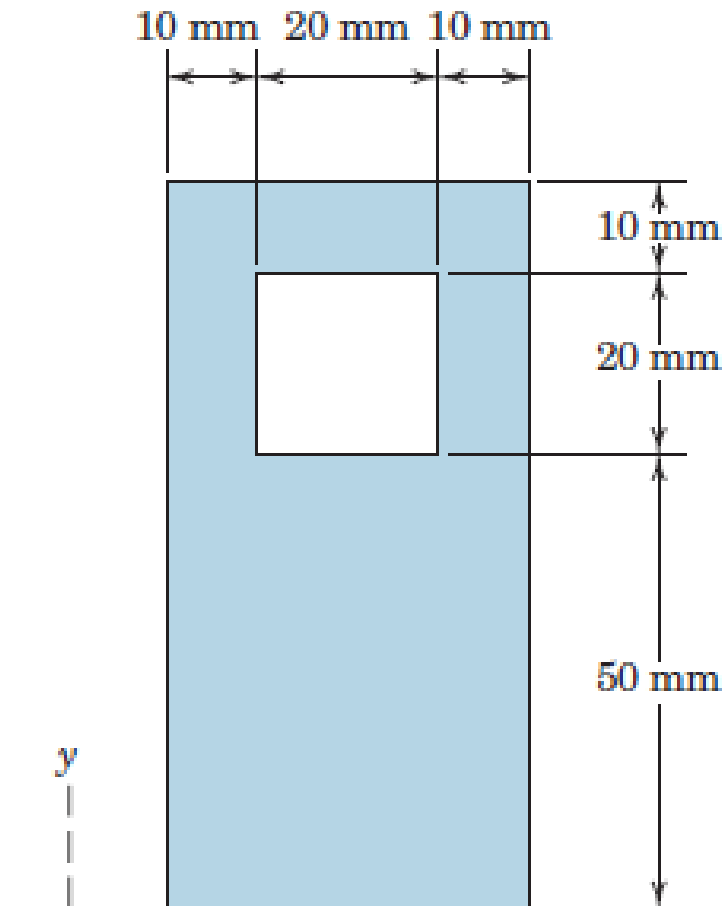
$$\bar{r} = \frac{2r}{\pi} = \frac{2(120)}{\pi} = 76.4 \text{ mm}$$

$$\bar{z} = r - \bar{r} = 120 - 76.4 = \underline{43.6 \text{ mm}}$$

$$\bar{x} = \underline{-120 \text{ mm}}, \text{ by inspection}$$

5/45 Determine the y -coordinate of the centroid of the shaded area.

Ans. $\bar{Y} = 37.1 \text{ mm}$

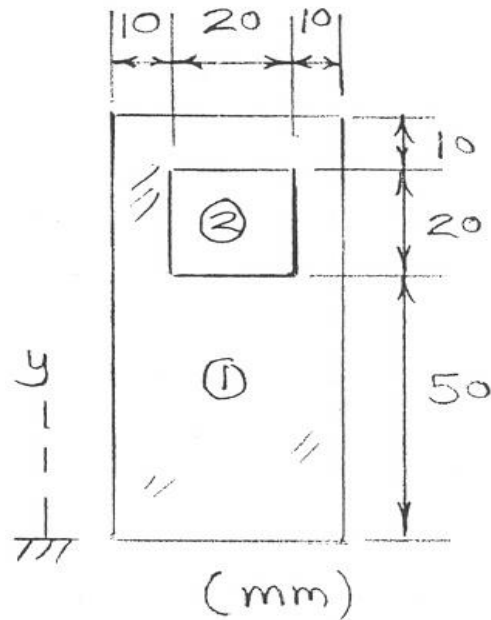


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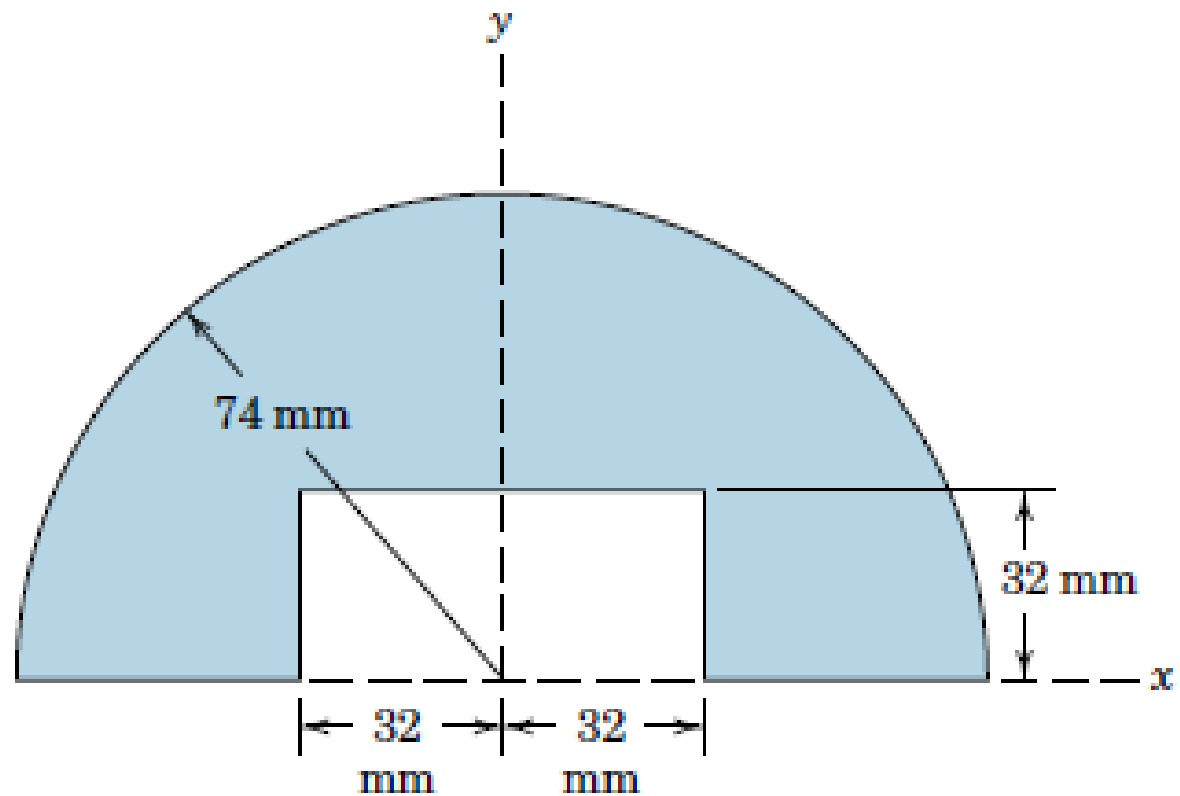
5/45



Component	$A \text{ (mm}^2\text{)}$	$\bar{y} \text{ (mm)}$	$\bar{y}A \text{ (mm}^3\text{)}$
1	$80(40) = 3200$	40	128 000
2	$-20(20) = -400$	60	-24 000
$\Sigma A = 2800$			$\Sigma \bar{y}A = 104 000$

$$\bar{Y} = \frac{\Sigma \bar{y}A}{\Sigma A} = \frac{104 000}{2800} = \underline{37.1 \text{ mm}}$$

5/46 Calculate the y -coordinate of the centroid of the shaded area.

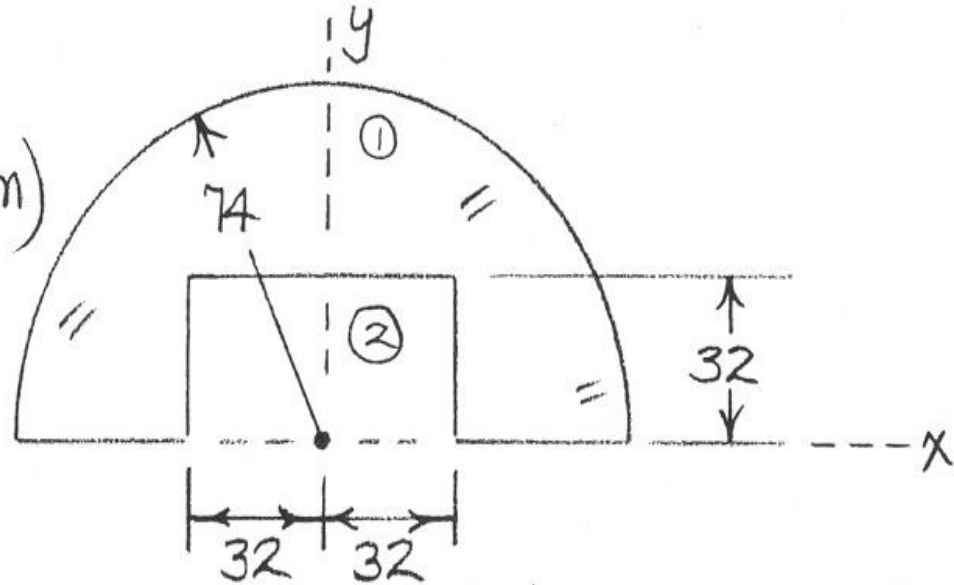


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5/46

(Dim. in mm)

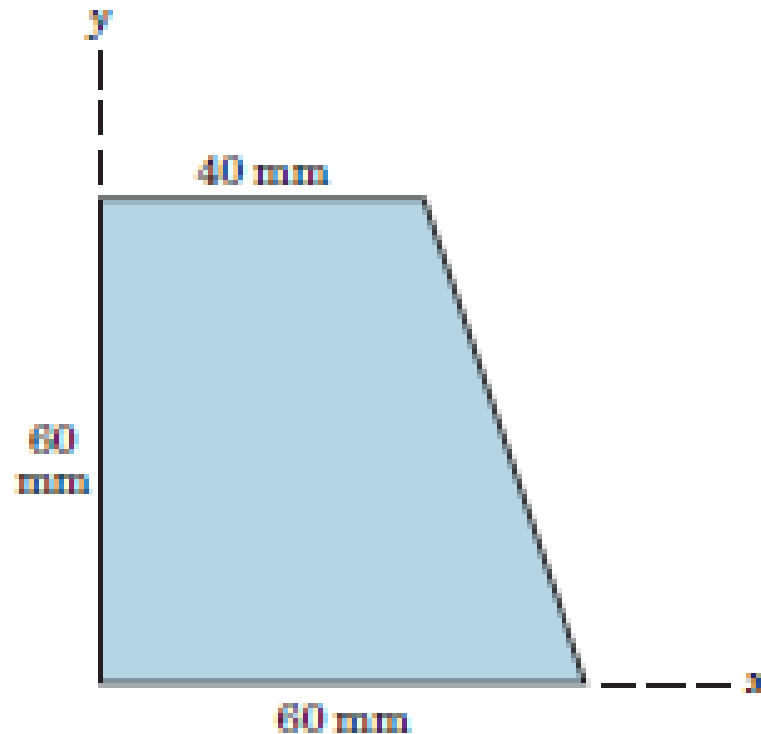


$$\bar{Y} = \frac{\sum A\bar{y}}{\sum A} = \frac{\pi \frac{74^2}{2} \left(\frac{4(74)}{3\pi} \right) - 64(32) \left(\frac{32}{2} \right)}{\pi \frac{74^2}{2} - 64(32)}$$

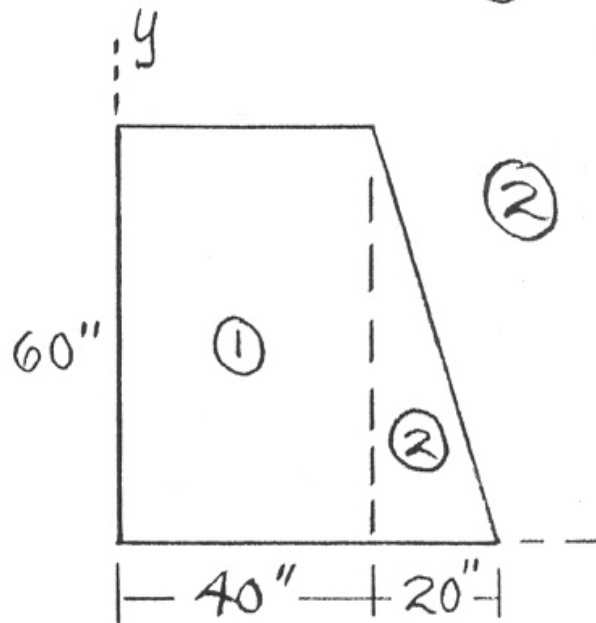
$$= \underline{36.2 \text{ mm}}$$

5/47 Determine the coordinates of the centroid of the trapezoidal area shown.

Ans. $\bar{X} = 25.3 \text{ mm}$, $\bar{Y} = 28.0 \text{ mm}$



5/47



$$\textcircled{1} A_1 = 40(60) = 2400 \text{ in.}^2$$

$$\bar{x}_1 = 20 \text{ in.}, \bar{y}_1 = 30 \text{ in.}$$

$$\textcircled{2} A_2 = \frac{1}{2}(20)(60) = 600 \text{ in.}^2$$

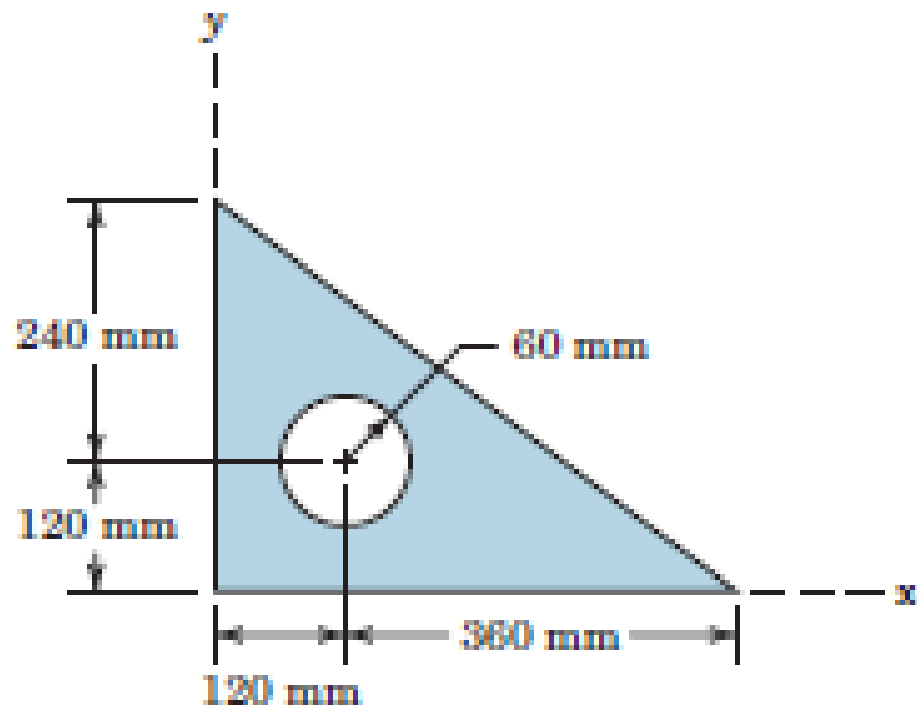
$$\bar{x}_2 = 40 + \frac{20}{3} = 46.7 \text{ in.}$$

$$\bar{y}_2 = \frac{60}{3} = 20 \text{ in.}$$

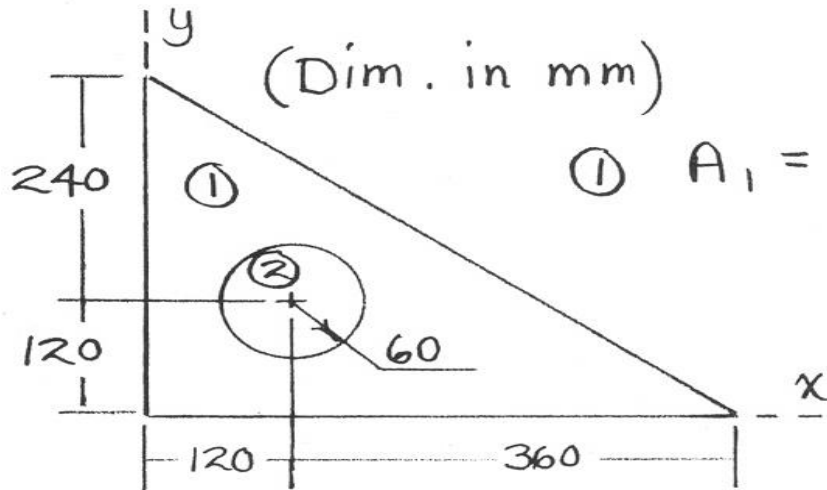
$$\bar{X} = \frac{\sum A \bar{x}}{\sum A} = \frac{2400(20) + 600(46.7)}{2400 + 600} = \underline{25.3 \text{ in.}}$$

$$\bar{Y} = \frac{\sum A \bar{y}}{\sum A} = \frac{2400(30) + 600(20)}{2400 + 600} = \underline{28.0 \text{ in.}}$$

5/48 Determine the x - and y -coordinates of the centroid of the shaded area.



5/48



$$\textcircled{1} A_1 = \frac{1}{2} (480)(360) = 86\,400 \text{ mm}^2$$

$$\bar{x}_1 = \frac{1}{3} (480) = 160 \text{ mm}$$

$$\bar{y}_1 = \frac{1}{3} (360) = 120 \text{ mm}$$

$$\textcircled{2} A_2 = -\pi (60)^2 = -11\,310 \text{ mm}^2$$

$$\bar{x}_2 = \bar{y}_2 = 120 \text{ mm}$$

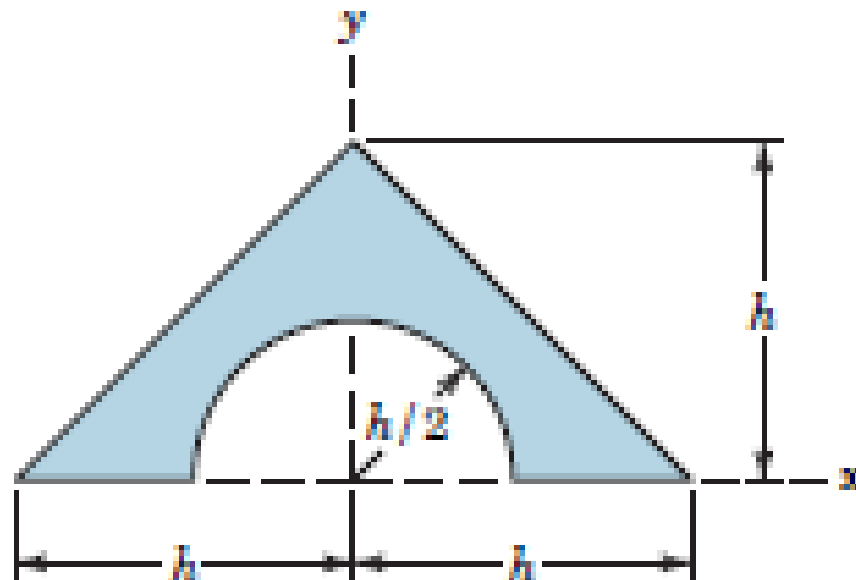
$$\bar{X} = \frac{\sum A \bar{x}}{\sum A} = \frac{86\,400 (160) - 11\,310 (120)}{86\,400 - 11\,310} = \underline{166.0 \text{ mm}}$$

$$\bar{Y} = \frac{\sum A \bar{y}}{\sum A} = \frac{86\,400 (120) - 11\,310 (120)}{86\,400 - 11\,310} = \underline{120 \text{ mm}}$$

(Because the circular hole is located at the $\bar{y}$ for the triangle, its presence does not alter the $\bar{y}$ result.)

5/49 Determine the y -coordinate of the centroid of the shaded area in terms of h .

Ans. $\bar{Y} = 0.412h$





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$$\boxed{5/49} \quad \text{Triangle: } A = \frac{1}{2}(2h)h = h^2$$

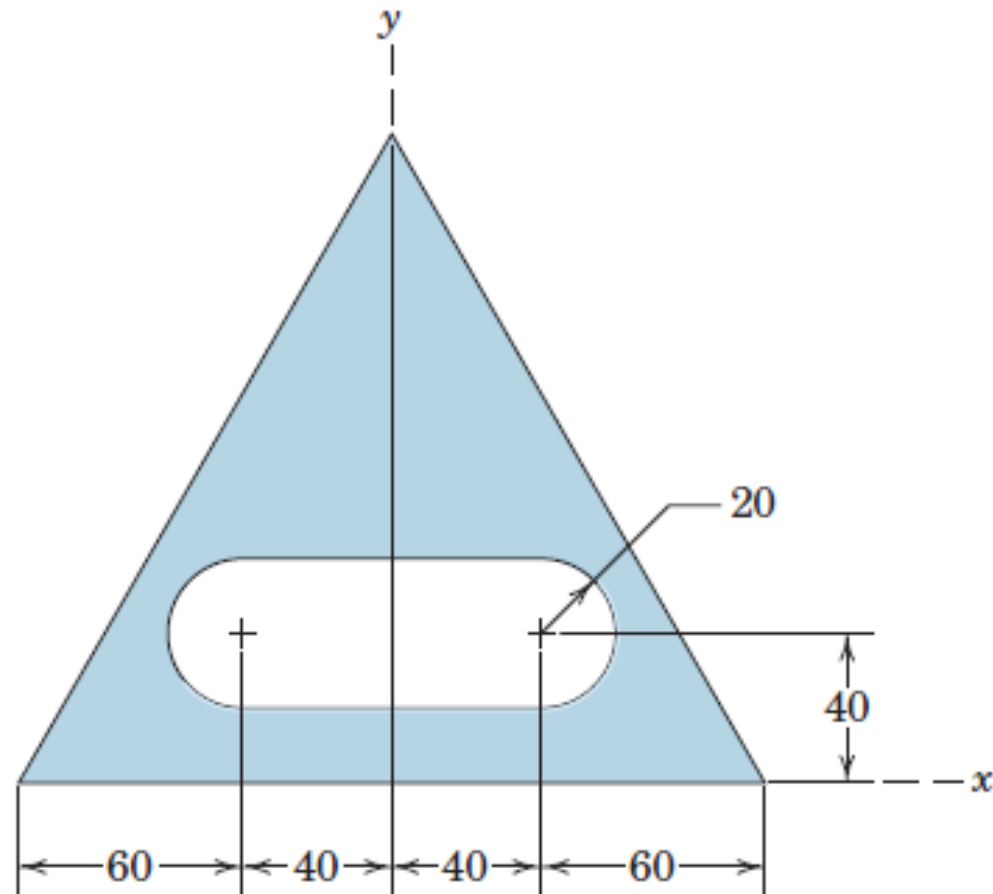
$$\bar{y} = h/3$$

$$\text{Semi-circular hole: } A = -\frac{1}{2}\pi(h/2)^2 = -\pi h^2/8$$

$$\bar{y} = 4(h/2)/3\pi = \frac{2h}{3\pi}$$

$$\bar{y} = \frac{\sum A\bar{y}}{\sum A} = \frac{h^2(h/3) - (\pi h^2/8)(2h/3\pi)}{h^2 - \pi h^2/8} = \frac{h}{4(1 - \pi/8)} = \underline{0.412h}$$

5/50 Determine the y -coordinate of the centroid of the shaded area. The triangle is equilateral.



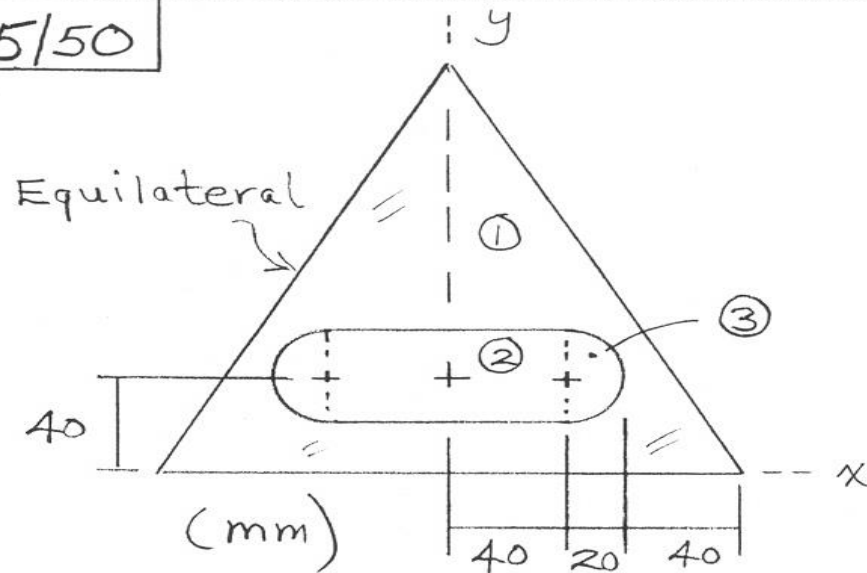
Dimensions in millimeters



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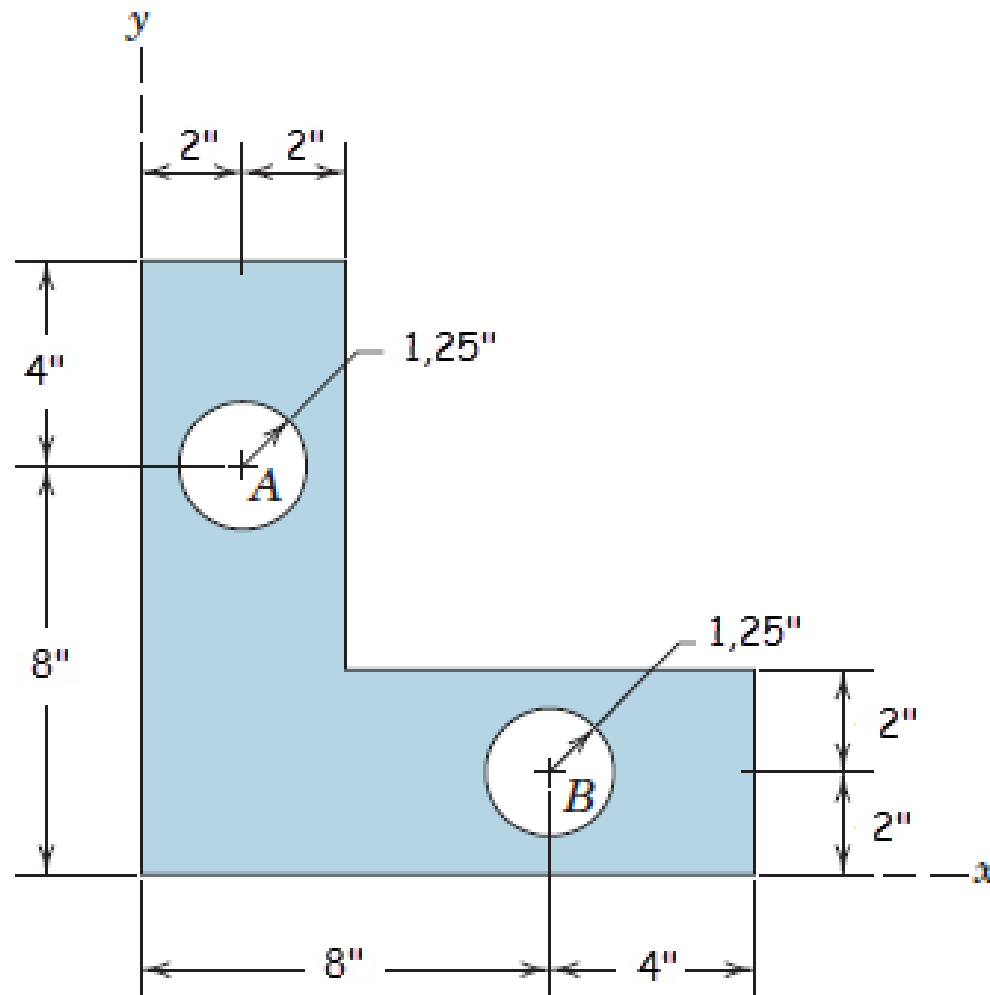
5/50



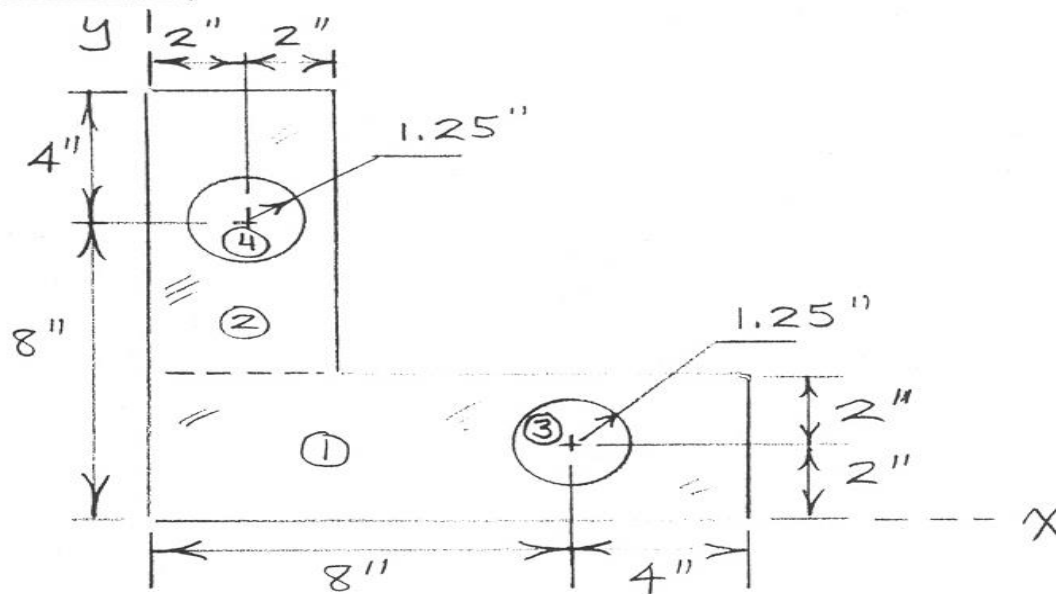
Component	$A \text{ (mm}^2\text{)}$	$\bar{y} \text{ (mm)}$	$\bar{y} A \text{ (mm}^3\text{)}$
Triangle 1	17 320	57.7	10^6
Rectangle 2	- 3 200	40	- 128 000
2 semicircles 3	- 1 257	40	- 50,300
$\Sigma A = 12 860$		$\Sigma \bar{y} A = 822 000$	

$$\bar{\bar{Y}} = \frac{\Sigma \bar{y} A}{\Sigma A} = \frac{822 000}{12 860} = 63.9 \text{ mm}$$

5/51 Determine the x - and y -coordinates of the centroid of the shaded area.



5/51



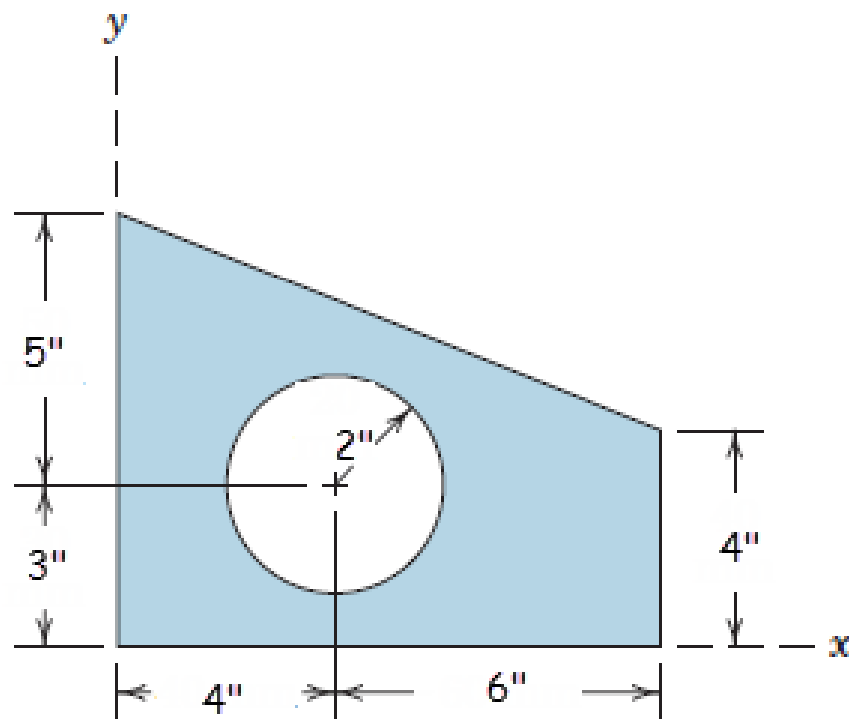
Comp.	A (in. ²)	$\bar{x}$ (in.)	$\bar{x}A$ (in. ³)
①	48	6	288
②	32	2	64
③	- 4.91	8	- 39.3
④	- 4.91	2	- 9.82

$$\Sigma A = 70.2$$

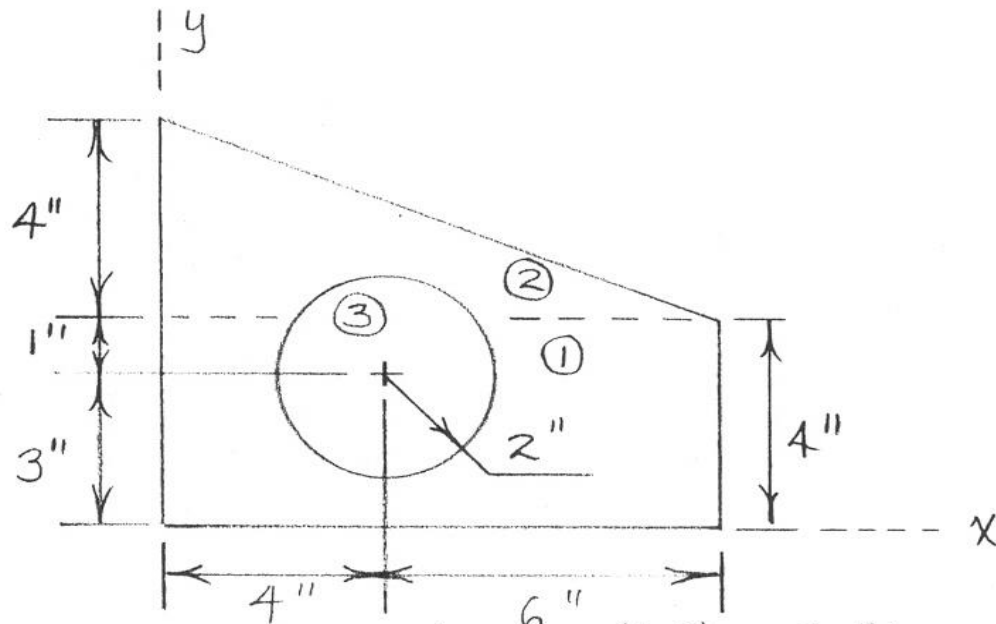
$$\Sigma \bar{x}A = 303$$

$$\bar{X} = \bar{Y} = \frac{\Sigma \bar{x}A}{\Sigma A} = \frac{303}{70.2} = \underline{4.32 \text{ in.}}$$

5/52 Determine the x - and y -coordinates of the centroid of the shaded area.



5/52



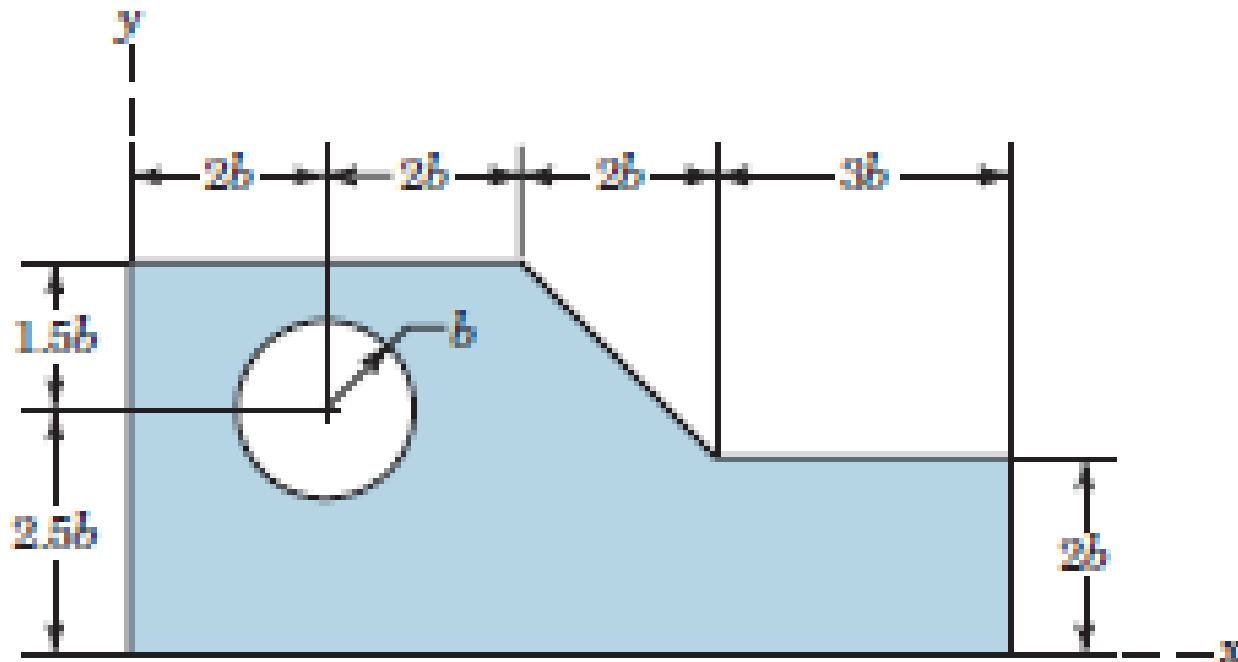
Part	(in. ²) A	(in.) $\bar{x}$	(in.) $\bar{y}$	(in. ³) $A\bar{x}$	(in. ³) $A\bar{y}$
1	40	5	2	200	80
2	20	$10/3$	$4 + 4/3$	66.7	106.7
3	$-\pi 2^2$	4	3	-50.3	-37.7
Totals	47.4			216	149.0

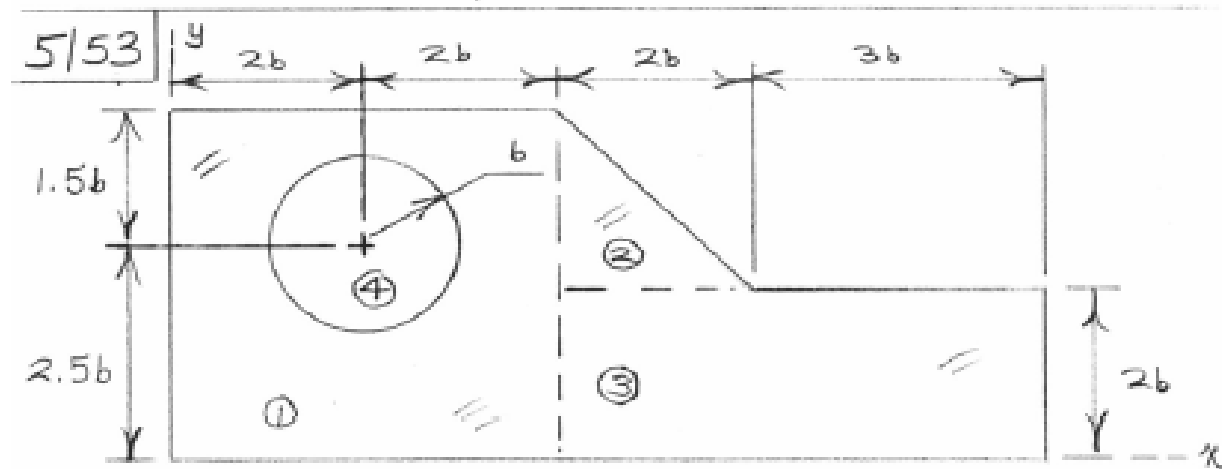
$$\bar{X} = \frac{\sum A\bar{x}}{\sum A} = \frac{216}{47.4} = \underline{4.56 \text{ in.}}$$

$$\bar{Y} = \frac{\sum A\bar{y}}{\sum A} = \frac{149.0}{47.4} = \underline{3.14 \text{ in.}}$$

5/53 Determine the x - and y -coordinates of the centroid of the shaded area.

Ans. $\bar{X} = 4.02b$, $\bar{Y} = 1.588b$





Comp.	A	$\bar{x}$	$\bar{y}$	$\bar{x}A$	$\bar{y}A$
1	$16b^2$	$2b$	$2b$	$32b^3$	$32b^3$
2	$2b^2$	$(4b + \frac{2b}{3})$	$(2b + \frac{2b}{3})$	$9.33b^3$	$5.33b^3$
3	$10b^2$	$(4b + \frac{5b}{2})$	b	$65b^3$	$10b^3$
4	$-\pi b^2$	$2b$	$2.5b$	$-2\pi b^3$	$-2.5\pi b^3$

$$\Sigma A = 24.9b^2$$

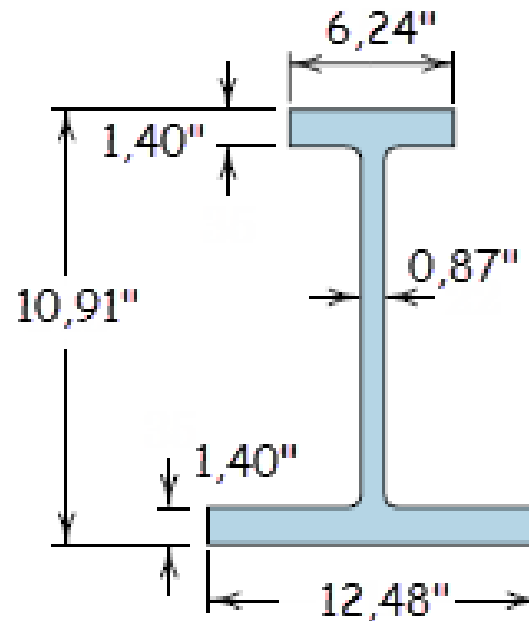
$$\Sigma \bar{x}A = 100.1b^3$$

$$\Sigma \bar{y}A = 39.5b^3$$

$$\bar{X} = \frac{\Sigma \bar{x}A}{\Sigma A} = \frac{100.1b^3}{24.9b^2} = 4.02b$$

$$\bar{Y} = \frac{\Sigma \bar{y}A}{\Sigma A} = \frac{39.5b^3}{24.9b^2} = 1.588b$$

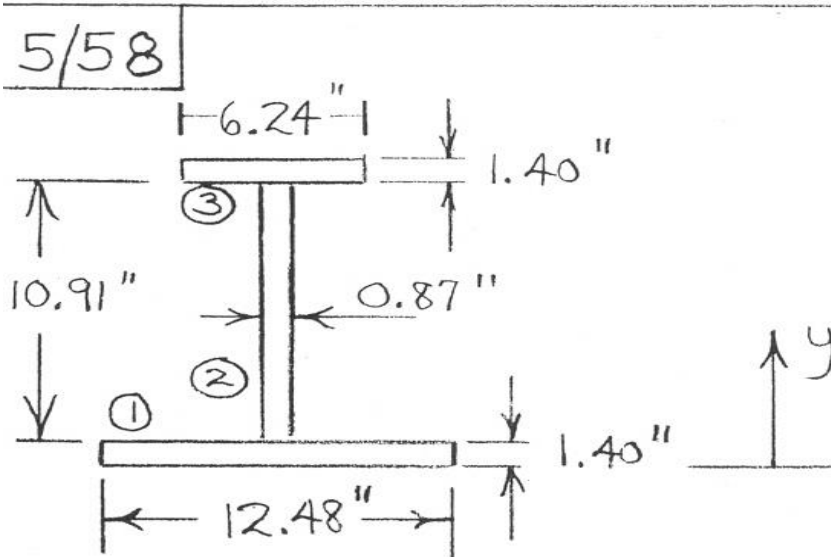
5/58 Determine the height above the base of the centroid of the cross-sectional area of the beam. Neglect the fillets.





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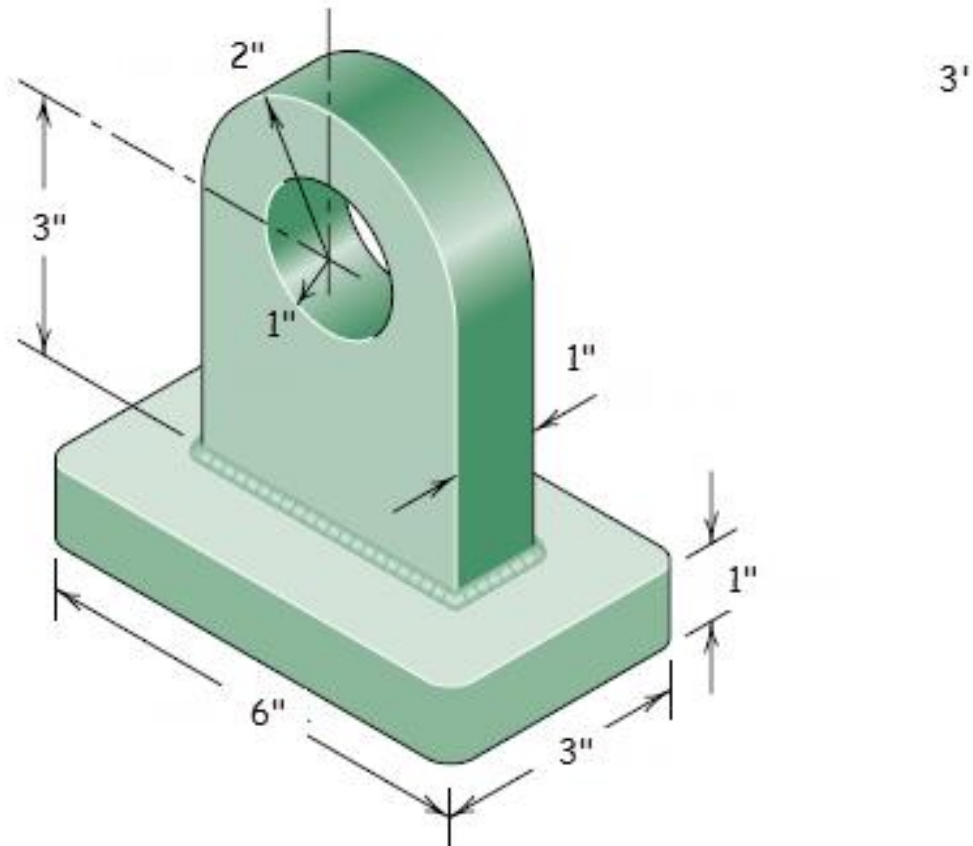
Comp.	$A \text{ (in.}^2\text{)}$	$\bar{y} \text{ (in.)}$	$A\bar{y} \text{ (in.}^3\text{)}$
①	$(12.48)(1.40)$	$\frac{1.40}{2}$	12.23
②	$(10.91)(0.87)$	$1.4 + \frac{10.91}{2}$	65.1
③	$(6.24)(1.40)$	$1.4 + 10.91 + \frac{1.4}{2}$	113.7

$$\Sigma A = 35.7$$

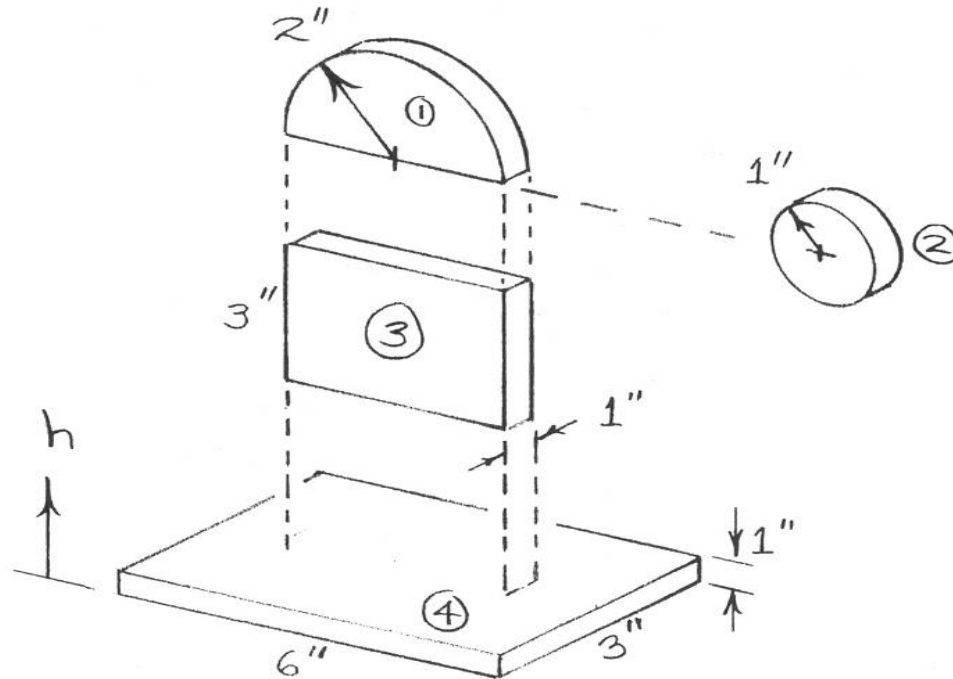
$$\Sigma A\bar{y} = 191.0$$

$$\bar{Y} = \frac{\Sigma A\bar{y}}{\Sigma A} = \underline{5.35 \text{ in.}}$$

5/66 Determine the distance $\bar{H}$ from the bottom of the base to the mass center of the bracket casting.



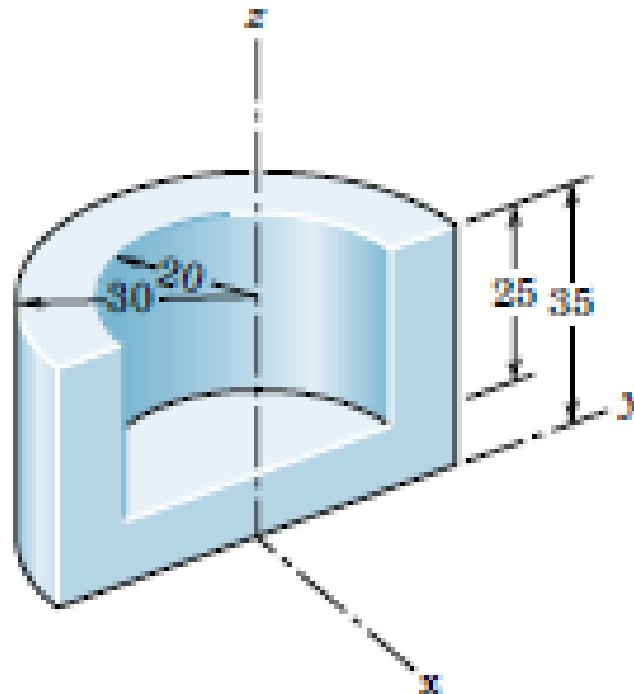
5/66



Part	$V, \text{in.}^3$	$\bar{h}, \text{in.}$	$V\bar{h}, \text{in.}^4$
①	6.28	4.85	30.5
②	-3.14	4	-12.57
③	12	2.5	30
④	18	0.5	9
Totals	33.1		56.9

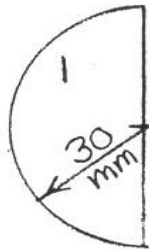
$$\bar{H} = \frac{\sum V\bar{h}}{\sum V} = \frac{56.9}{33.1} = \underline{1.717 \text{ in.}}$$

5/68 Calculate the coordinates of the mass center of the metal die casting shown.



Dimensions in millimeters

5/68



Length = 35 mm



Length = 25 mm

$$\begin{cases} V_1 = \frac{\pi}{2} (30)^2 (35) = 49\,500 \text{ mm}^3 \\ \bar{x}_1 = -\frac{4(30)}{3\pi} = -12.73 \text{ mm} \\ \bar{z}_1 = 17.5 \text{ mm} \end{cases}$$

$$V_2 = -\frac{\pi}{2} (20)^2 (25) = -15\,710 \text{ mm}^3$$

$$\bar{x}_2 = -\frac{4(20)}{3\pi} = -8.49 \text{ mm}$$

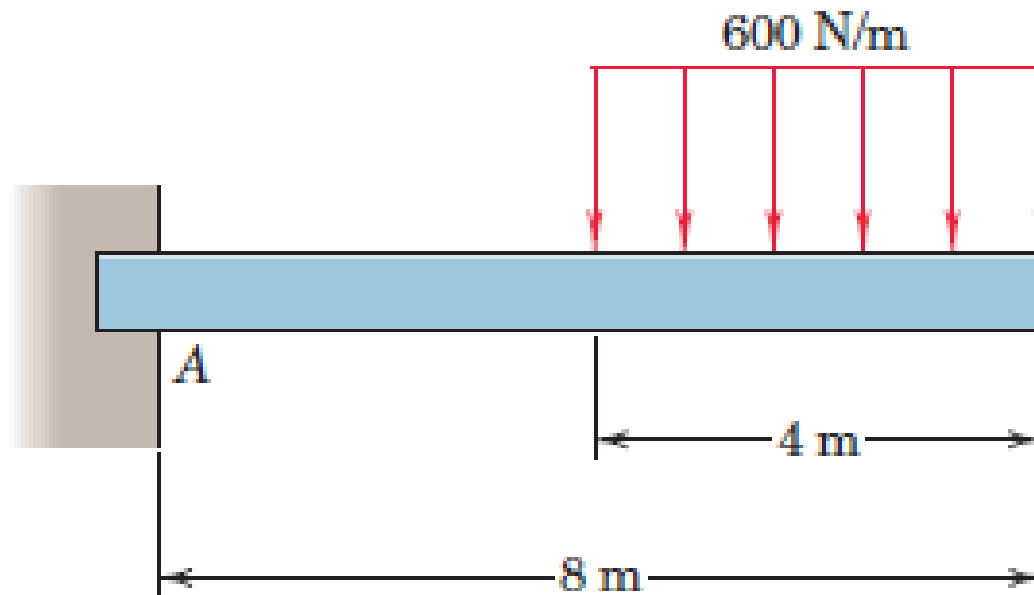
$$\bar{z}_2 = \frac{1}{2}(10 + 35) = 22.5 \text{ mm}$$

$$\begin{aligned} \bar{X} &= \frac{\sum V \bar{x}}{\sum V} = \frac{49\,500(-12.73) - 15\,710(-8.49)}{49\,500 - 15\,710} \\ &= \underline{-14.71 \text{ mm}} \end{aligned}$$

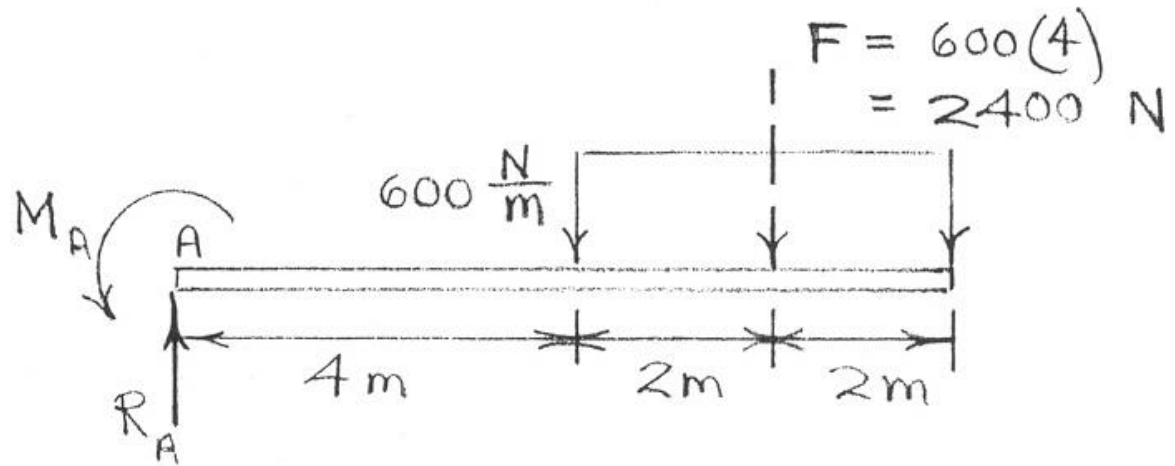
$$\begin{aligned} \bar{Z}_1 &= \frac{\sum V \bar{z}}{\sum V} = \frac{49\,500(17.5) - 15\,710(22.5)}{49\,500 - 15\,710} \\ &= \underline{15.17 \text{ mm}} \end{aligned}$$

5/95 Calculate the supporting force R_A and moment M_A at A for the loaded cantilever beam.

Ans. $R_A = 2.4 \text{ kN}$, $M_A = 14.4 \text{ kN} \cdot \text{m}$ CCW



5/95



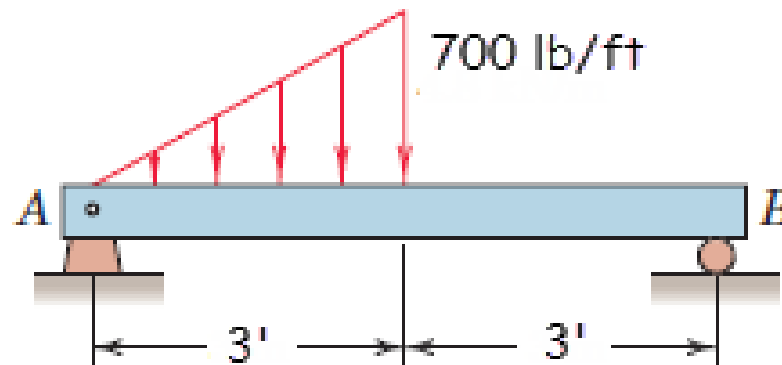
$$+\uparrow \sum F = 0 : R_A - 2400 = 0 ,$$

$$\underline{R_A = 2.4 \text{ kN}}$$

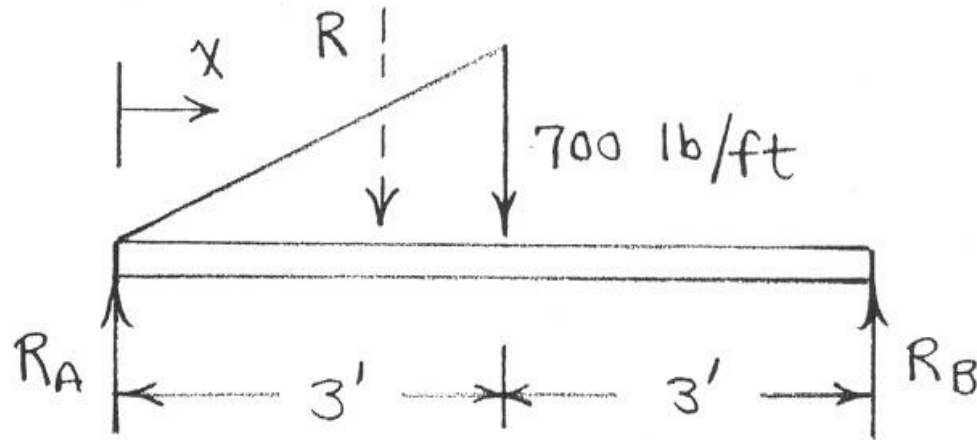
$$+\circlearrowleft \sum M_A = 0 : M_A - 2400(6) = 0 ,$$

$$\underline{M_A = 14.4 \text{ kN}\cdot\text{m}}$$

5/96 Calculate the reactions at A and B for the beam subjected to the triangular load distribution.



5/96



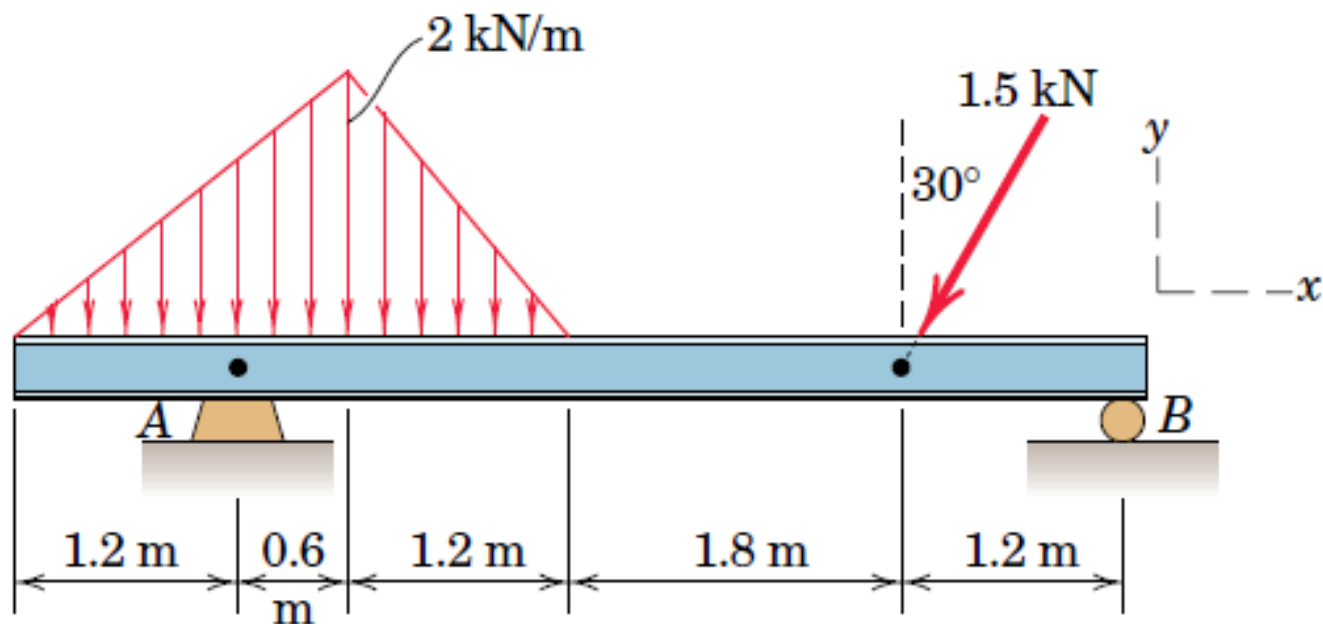
$$R = \frac{1}{2} (700)(6) = 1050 \text{ lb} \quad @ \quad \bar{x} = \frac{2}{3}(6) = 2'$$

$$\curvearrowright \sum M_A = 0 : R_B (6) - 1050 (2) = 0, \quad \underline{R_B = 350 \text{ lb}}$$

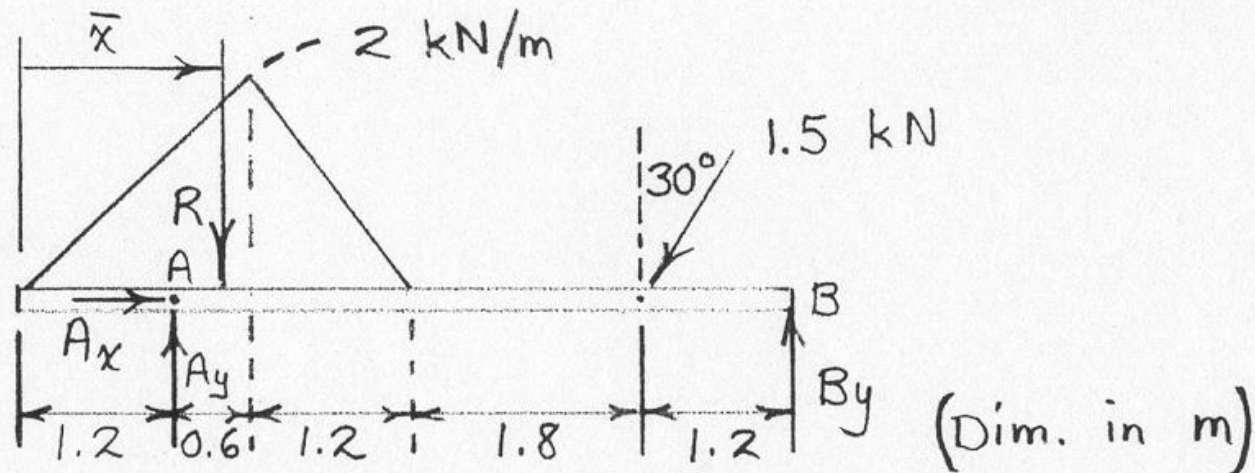
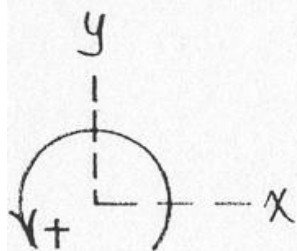
$$+\uparrow \sum F = 0 : R_A + 350 - 1050 = 0, \quad \underline{R_A = 700 \text{ lb}}$$

5/101 Determine the reactions at A and B for the beam subjected to a combination of distributed and point loads.

Ans. $A_x = 750 \text{ N}$, $A_y = 3.07 \text{ kN}$, $B_y = 1.224 \text{ kN}$



5/101



$$\begin{cases} R = \frac{1}{2} (1.2 + 0.6 + 1.2) (2) = 3 \text{ kN} \\ \bar{x} = \frac{(1.2 + 0.6) + (1.2 + 0.6 + 1.2)}{3} = 1.6 \text{ m} \end{cases}$$

$$\sum F_x = 0 : A_x - 1.5 \sin 30^\circ = 0, \quad \underline{A_x = 750 \text{ N}}$$

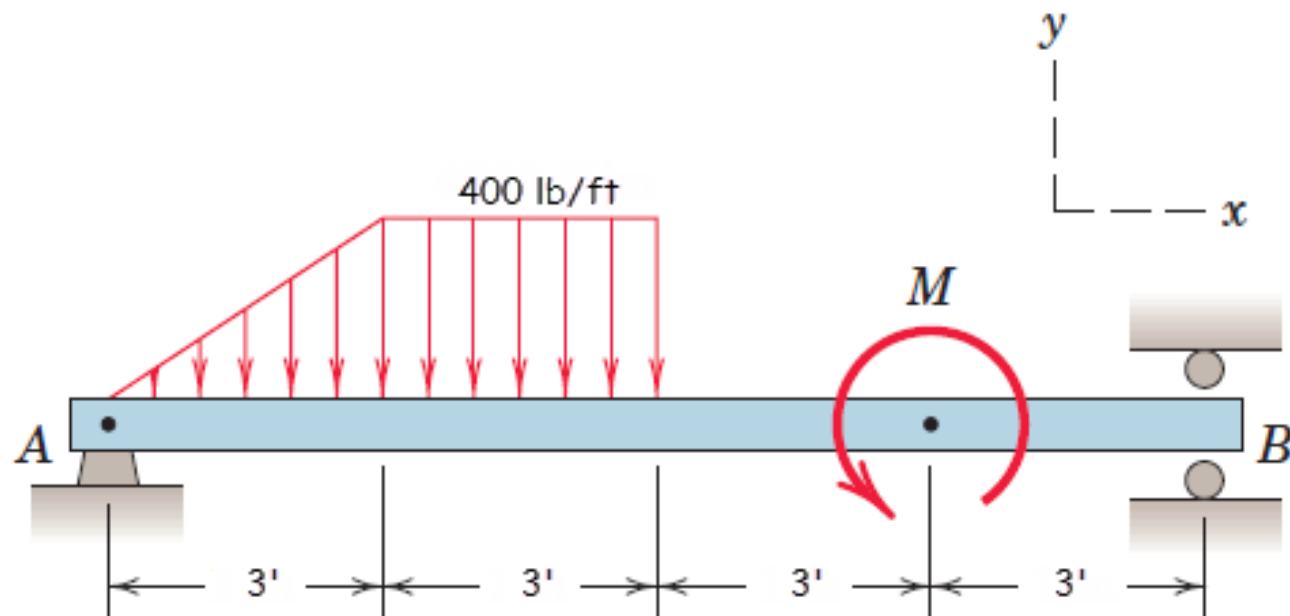
$$\sum M_A = 0 : -3 (1.6 - 1.2) - 1.5 \cos 30^\circ (3.6) - B_y (4.8) = 0$$

$$\underline{B_y = 1.224 \text{ kN}}$$

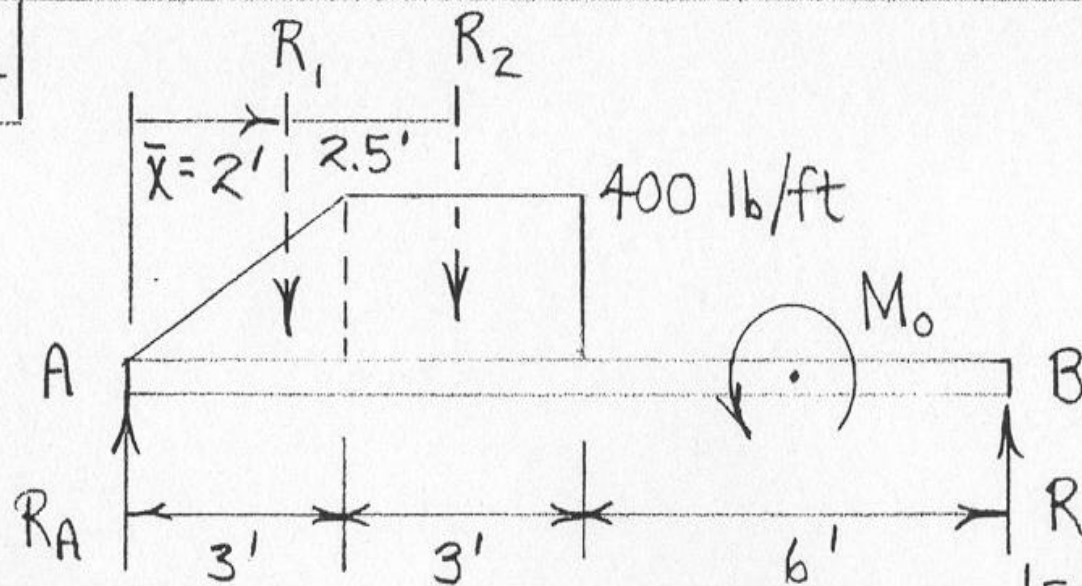
$$\sum F_y = 0 : A_y - 3 - 1.5 \cos 30^\circ + 1.224 = 0$$

$$\underline{A_y = 3.07 \text{ kN}}$$

5/104 The beam is subjected to the distributed load and the couple shown. If M is slowly increased starting from zero, at what value M_0 will contact at B change from the lower surface to the upper surface?



5/104



$$R_1 = \frac{1}{2} (400) (3) = 600 \text{ lb @ } \bar{x} = \frac{2}{3}(3) = 2'$$

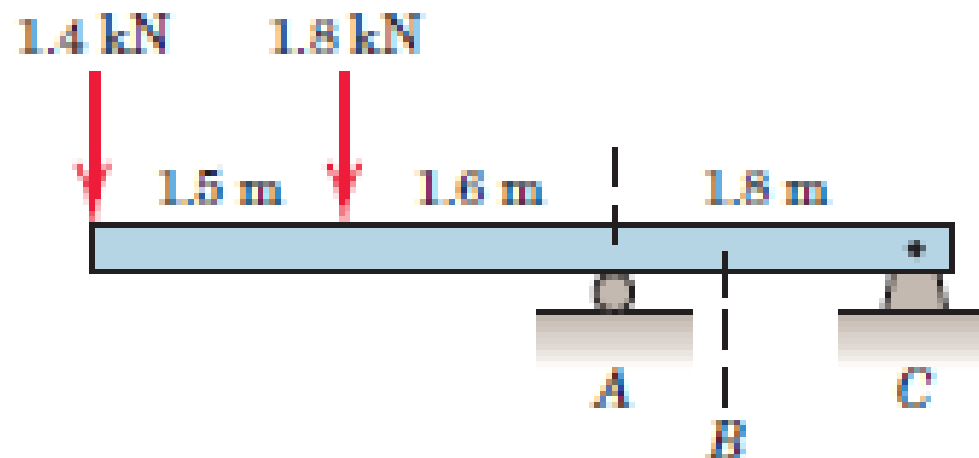
$$R_2 = 400 (3) = 1200 \text{ lb}$$

$R_B = 0$ at
switch from
lower to upper
surface

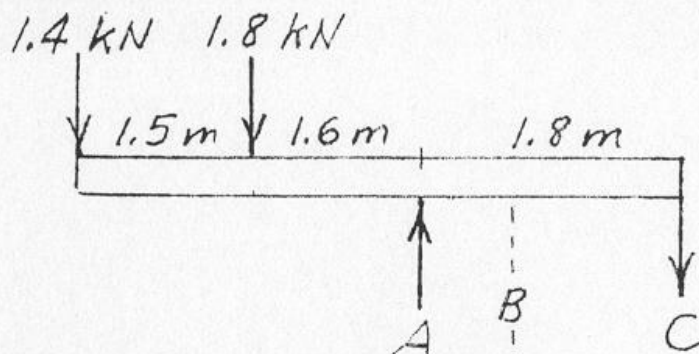
$$\uparrow + \sum M_A = 0 : -600(2) - 1200(4.5) + M_0 = 0$$

$$\underline{M_0 = 6600 \text{ lb-ft}}$$

5/118 Determine the shear V at a section B between A and C and the moment M at the support A .



5/118



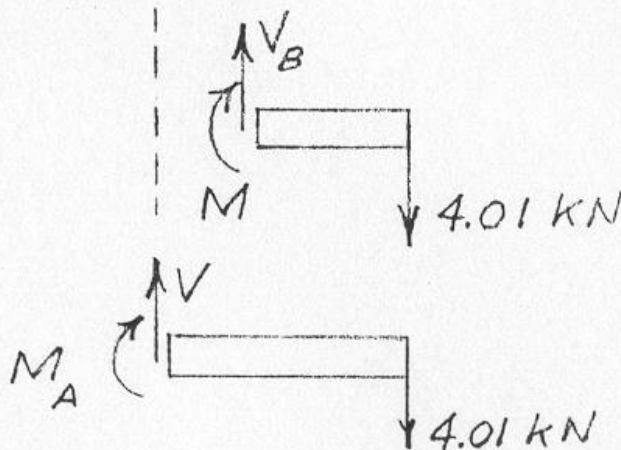
$$\sum M_A = 0;$$

$$1.4(3.1) + 1.8(1.6) - 1.8C = 0$$

$$C = 4.01 \text{ kN}$$

$$\sum F = 0; A - 4.01 - 1.4 - 1.8 = 0$$

$$A = 7.21 \text{ kN}$$



$$\sum F = 0; \underline{V_B = 4.01 \text{ kN}}$$

$$\sum M_A = 0; M_A + 4.01(1.8) = 0$$

$$\underline{M_A = -7.22 \text{ kN}\cdot\text{m}}$$

Problem 403

Beam loaded as shown in Fig.403. Draw Shear force and Bending Moment diagrams.

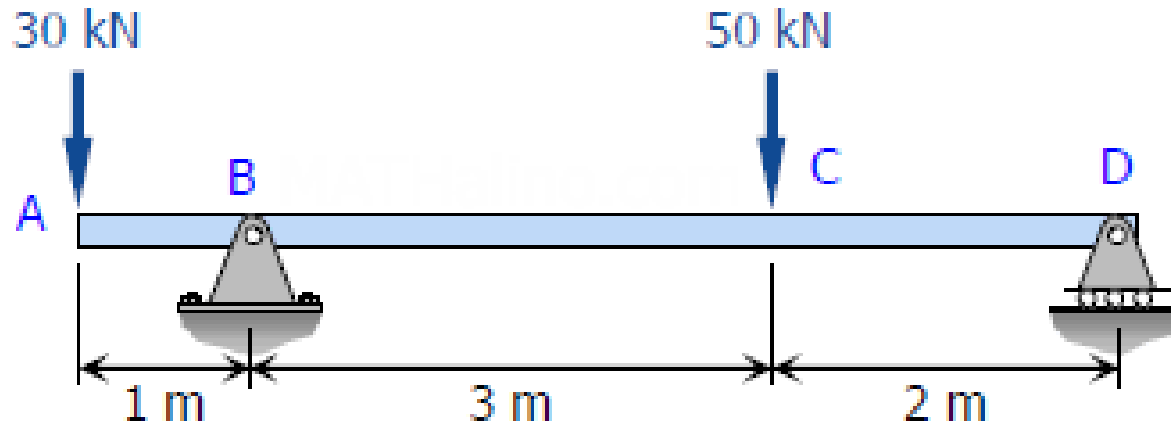


Figure P-403

From the load diagram:

$$\Sigma M_B = 0$$

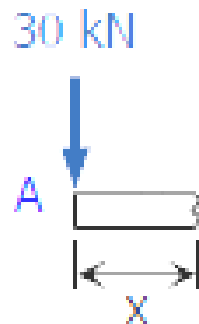
$$5R_D + 1(30) = 3(50)$$

$$R_D = 24 \text{ kN}$$

$$\Sigma M_D = 0$$

$$5R_B = 2(50) + 6(30)$$

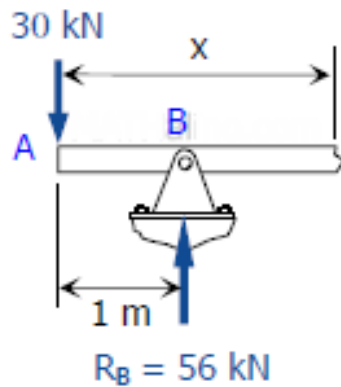
$$R_B = 56 \text{ kN}$$



Segment AB:

$$V_{AB} = -30 \text{ kN}$$

$$M_{AB} = -30x \text{ kN} \cdot \text{m}$$



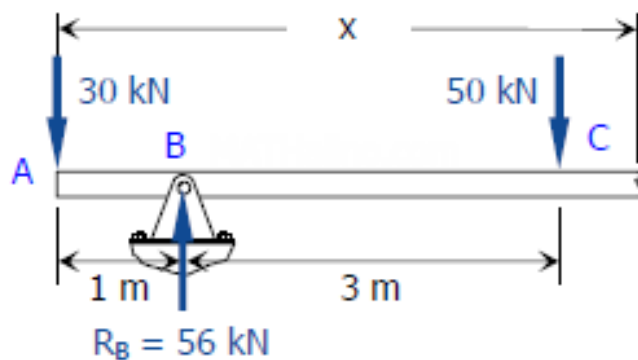
Segment BC:

$$V_{BC} = -30 + 56$$

$$V_{BC} = 26 \text{ kN}$$

$$M_{BC} = -30x + 56(x - 1)$$

$$M_{BC} = 26x - 56 \text{ kN} \cdot \text{m}$$



Segment CD:

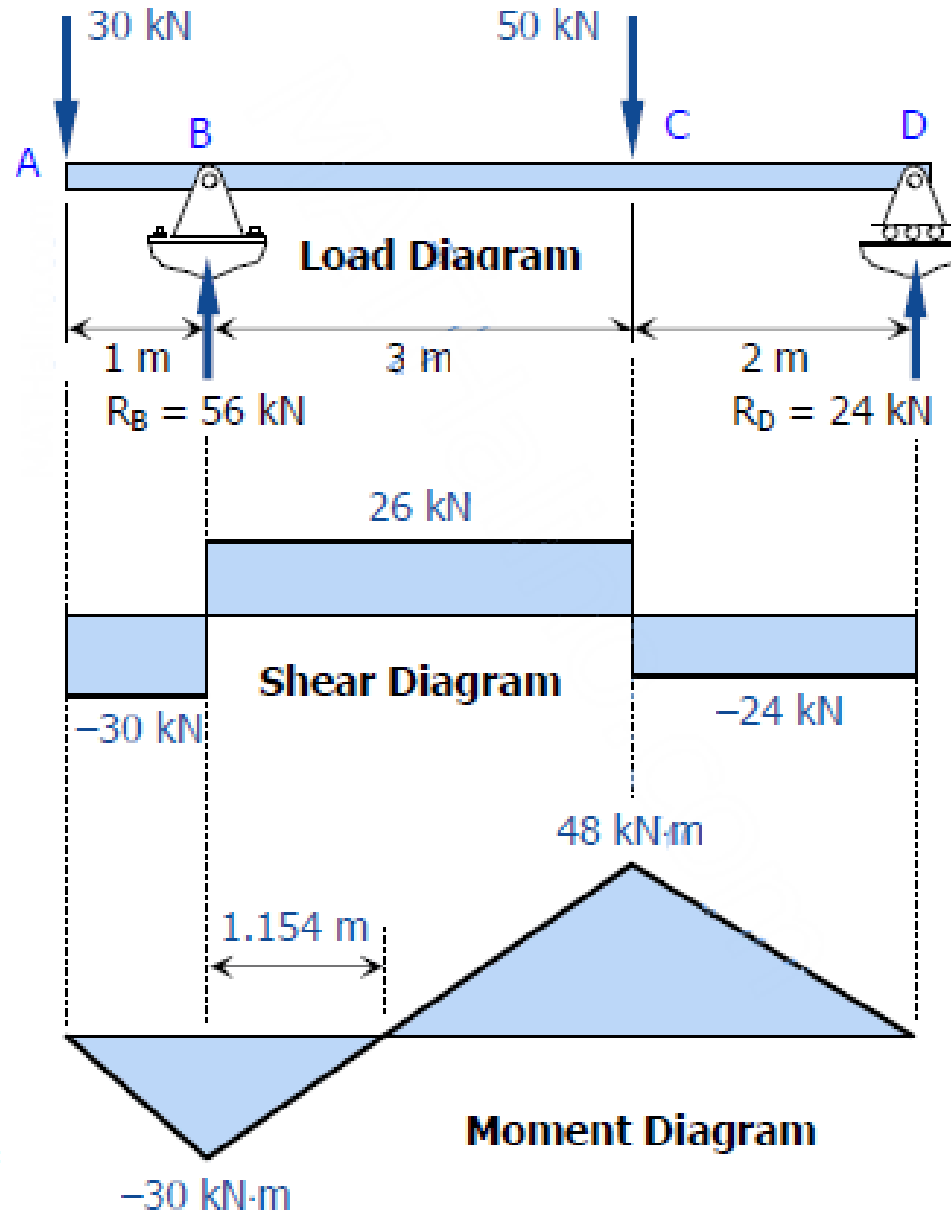
$$V_{CD} = -30 + 56 - 50$$

$$V_{CD} = -24 \text{ kN}$$

$$M_{CD} = -30x + 56(x - 1) - 50(x - 4)$$

$$M_{CD} = -30x + 56x - 56 - 50x + 200$$

$$M_{CD} = -24x + 144 \text{ kN} \cdot \text{m}$$



Problem 404

Beam loaded as shown in Fig. P-404. Draw Shear Force and Bending Moment diagrams.

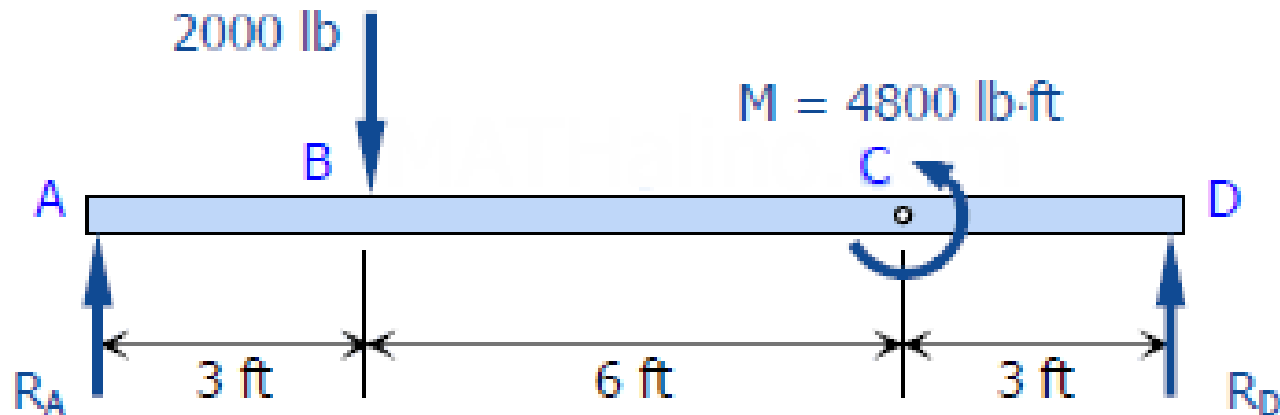


Figure P-404

$$\Sigma M_A = 0$$

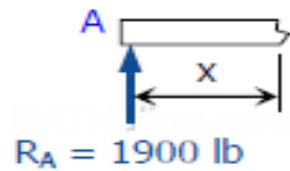
$$12R_D + 4800 = 3(2000)$$

$$R_D = 100 \text{ lb}$$

$$\Sigma M_D = 0$$

$$12R_A = 9(2000) + 4800$$

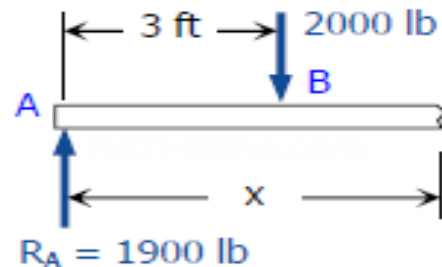
$$R_A = 1900 \text{ lb}$$



Segment AB:

$$V_{AB} = 1900 \text{ lb}$$

$$M_{AB} = 1900x \text{ lb} \cdot \text{ft}$$



Segment BC:

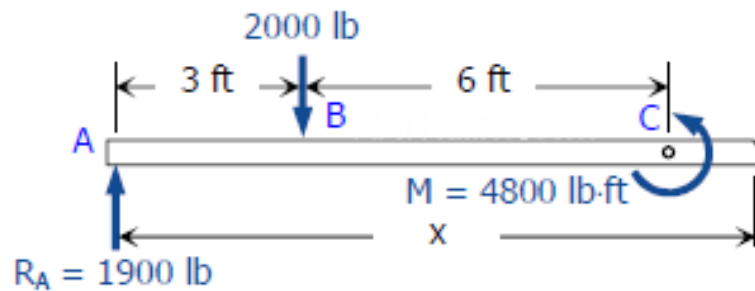
$$V_{BC} = 1900 - 2000$$

$$V_{BC} = -100 \text{ lb}$$

$$M_{BC} = 1900x - 2000(x - 3)$$

$$M_{BC} = 1900x - 2000x + 6000$$

$$M_{BC} = -100x + 6000 \text{ lb} \cdot \text{ft}$$



Segment CD:

$$V_{CD} = 1900 - 2000$$

$$V_{CD} = -100 \text{ lb}$$

$$M_{CD} = 1900x - 2000(x - 3) - 4800$$

$$M_{CD} = 1900x - 2000x + 6000 - 4800$$

$$M_{CD} = -100x + 1200 \text{ lb} \cdot \text{ft}$$

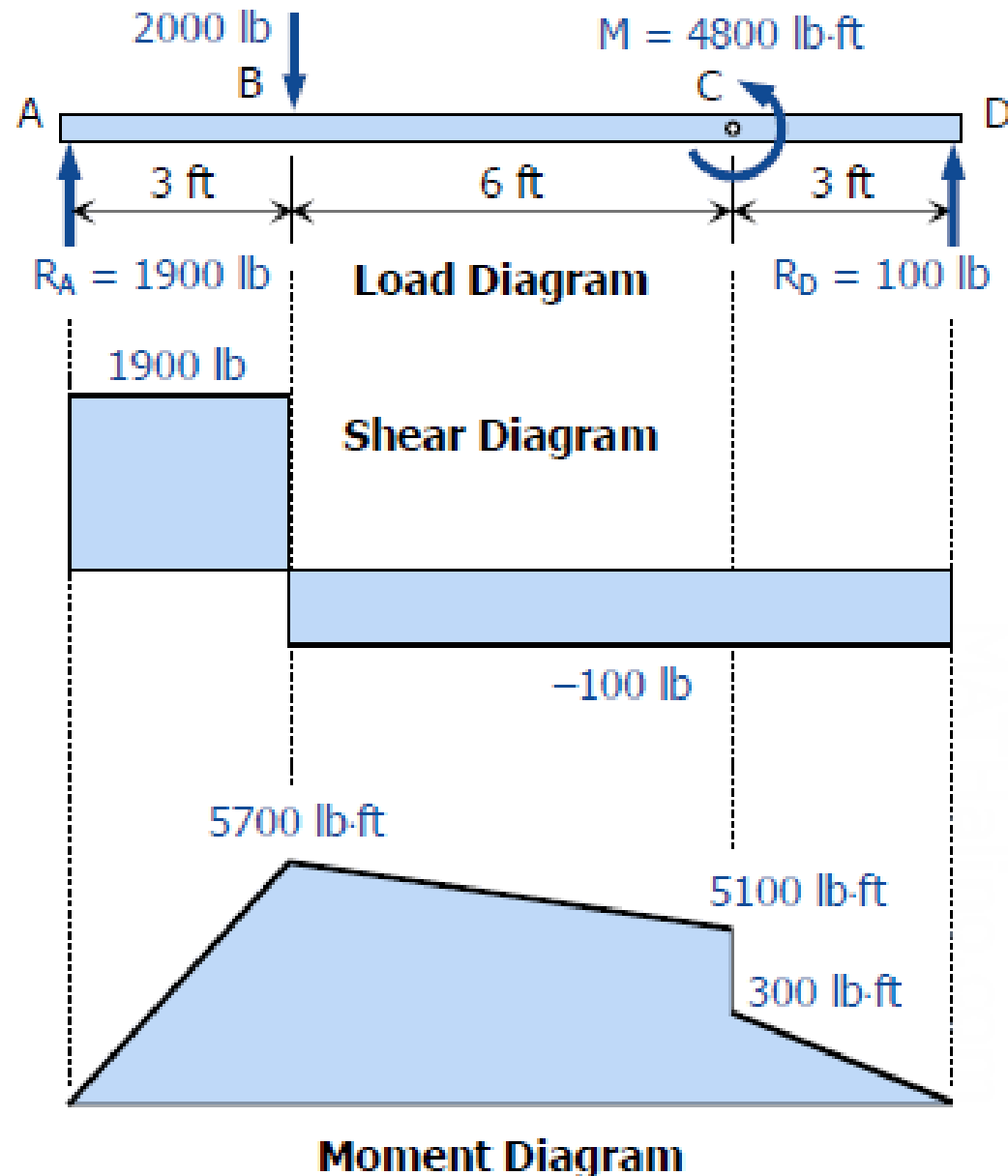
To draw the shear diagram:

1. At segment AB, the shear is uniformly distributed at 1900 lb.
2. A shear of -100 lb is uniformly distributed over segments BC and CD.

To draw the Moment Diagram:

1. $M_{AB} = 1900x$ is linear; at $x = 0$, $M_{AB} = 0$; at $x = 3$ ft, $M_{AB} = 5700$ lb·ft.
2. For segment BC, $M_{BC} = -100x + 6000$ is linear; at $x = 3$ ft, $M_{BC} = 5700$ lb·ft; at $x = 9$ ft, $M_{BC} = 5100$ lb·ft.

3. $M_{CD} = -100x + 1200$ is again linear; at $x = 9$ ft, $M_{CD} = 300$ lb·ft; at $x = 12$ ft, $M_{CD} = 0$.



Problem 405

Beam loaded as shown in Fig. P-405. Draw Shear force and Bending Moment diagrams.

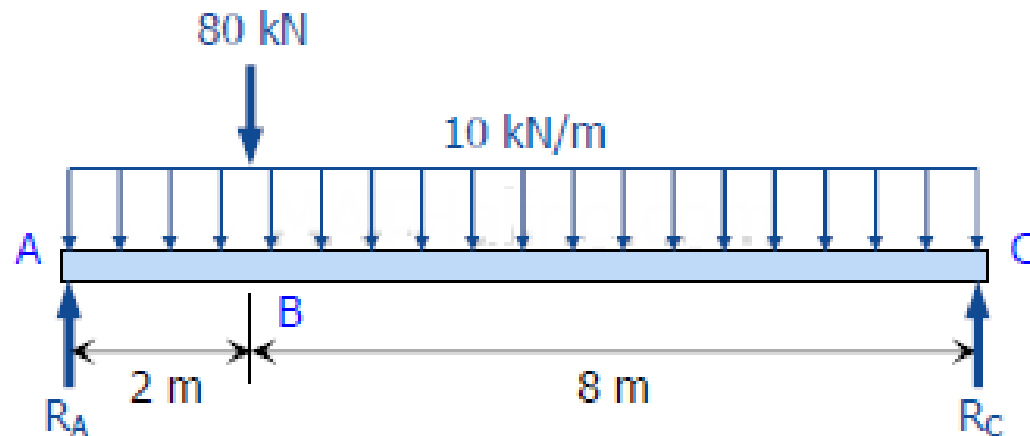


Figure P-405

$$\Sigma M_A = 0$$

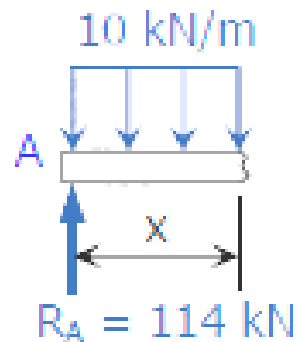
$$10R_C = 2(80) + 5[10(10)]$$

$$R_C = 66 \text{ kN}$$

$$\Sigma M_C = 0$$

$$10R_A = 8(80) + 5[10(10)]$$

$$R_A = 114 \text{ kN}$$



Segment AB:

$$V_{AB} = 114 - 10x \text{ kN}$$

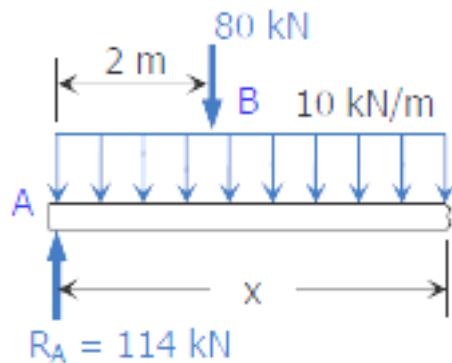
$$M_{AB} = 114x - 10x(x/2)$$

$$M_{AB} = 114x - 5x^2 \text{ kN} \cdot \text{m}$$



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Segment BC:

$$V_{BC} = 114 - 80 - 10x$$

$$V_{BC} = 34 - 10x \text{ kN}$$

$$M_{BC} = 114x - 80(x - 2) - 10x(x/2)$$

$$M_{BC} = 160 + 34x - 5x^2 \text{ kN} \cdot \text{m}$$

To draw the Shear Diagram:

1. For segment AB, $V_{AB} = 114 - 10x$ is linear; at $x = 0$, $V_{AB} = 114 \text{ kN}$; at $x = 2 \text{ m}$, $V_{AB} = 94 \text{ kN}$.
2. $V_{BC} = 34 - 10x$ for segment BC is linear; at $x = 2 \text{ m}$, $V_{BC} = 14 \text{ kN}$; at $x = 10 \text{ m}$, $V_{BC} = -66 \text{ kN}$. When $V_{BC} = 0$, $x = 3.4 \text{ m}$ thus $V_{BC} = 0$ at 1.4 m from B.

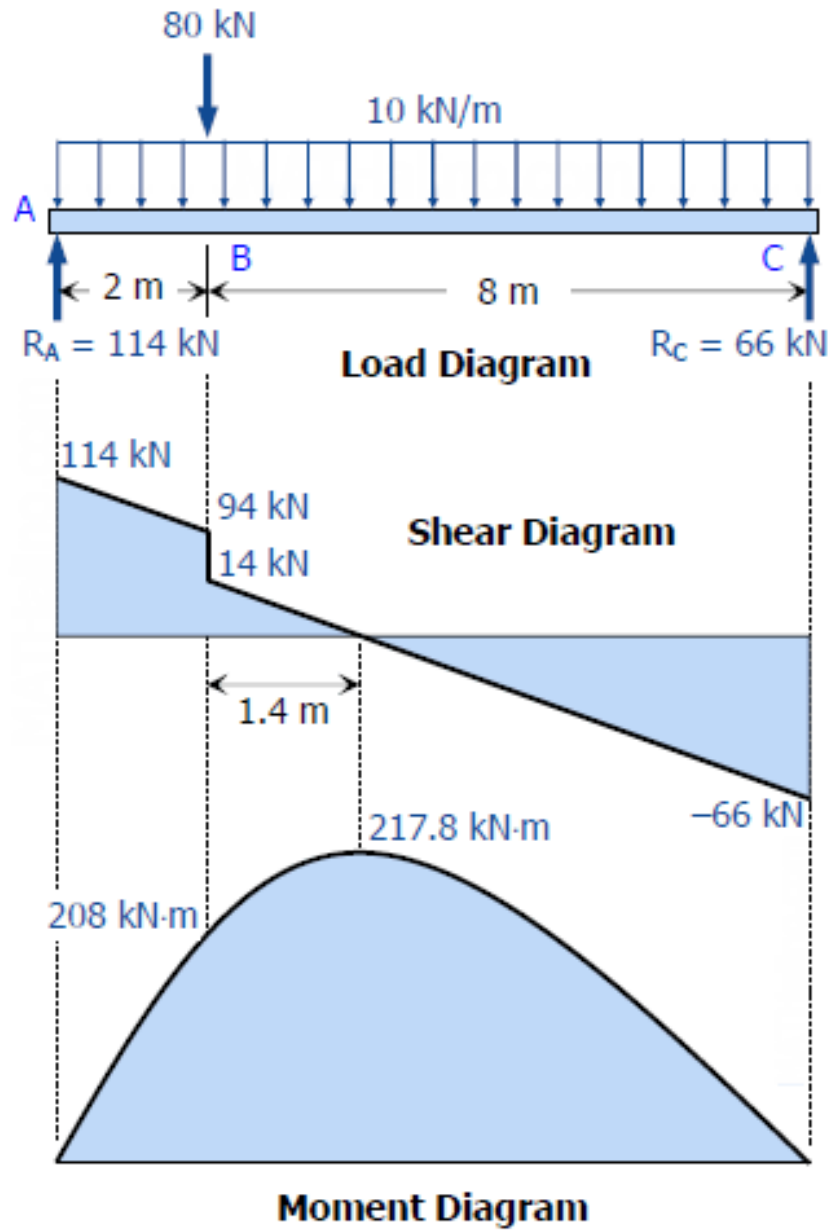


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To draw the Moment Diagram:

1. $M_{AB} = 114x - 5x^2$ is a second degree curve for segment AB; at $x = 0$, $M_{AB} = 0$; at $x = 2$ m, $M_{AB} = 208$ kN·m.
2. The moment diagram is also a second degree curve for segment BC given by $M_{BC} = 160 + 34x - 5x^2$; at $x = 2$ m, $M_{BC} = 208$ kN·m; at $x = 10$ m, $M_{BC} = 0$.
3. Note that the maximum moment occurs at point of zero shear. Thus, at $x = 3.4$ m, $M_{BC} = 217.8$ kN·m.





Relationship Between Load, Shear, and Moment

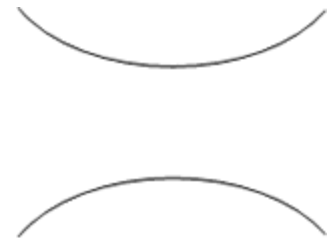
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Properties of Shear and Moment Diagrams

The following are some important properties of shear and moment diagrams:

1. The area of the shear diagram to the left or to the right of the section is equal to the moment at that section.
2. The slope of the moment diagram at a given point is the shear at that point.
3. The slope of the shear diagram at a given point equals the load at that point.
4. The maximum moment occurs at the point of zero shears. This is in reference to property number 2, that when the shear (also the slope of the moment diagram) is zero, the tangent drawn to the moment diagram is horizontal.
5. When the shear diagram is increasing, the moment diagram is concave upward.
6. When the shear diagram is decreasing, the moment diagram is concave downward.



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Sign Convention

The customary sign conventions for shearing force and bending moment are represented by the figures below.

A force that tends to bend the beam downward is said to produce a positive bending moment.

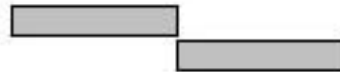
A force that tends to shear the left portion of the beam upward with respect to the right portion is said to produce a positive shearing force.



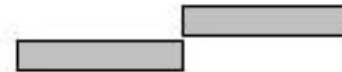
Positive Bending



Negative Bending



Positive Shear



Negative Shear

An easier way of determining the sign of the bending moment at any section is that upward forces always cause positive bending moments regardless of whether they act to the left or to the right of the exploratory section.

Problem 425

Beam loaded as shown in Fig. P-425.

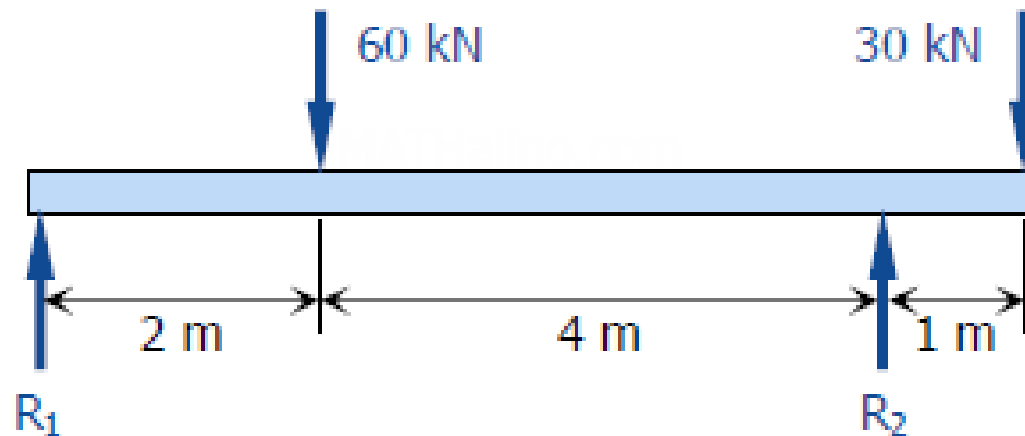


Figure P-425

$$\Sigma M_A = 0$$

$$6R_2 = 2(60) + 7(30)$$

$$R_2 = 55 \text{ kN}$$

$$\Sigma M_C = 0$$

$$6R_1 + 1(30) = 4(60)$$

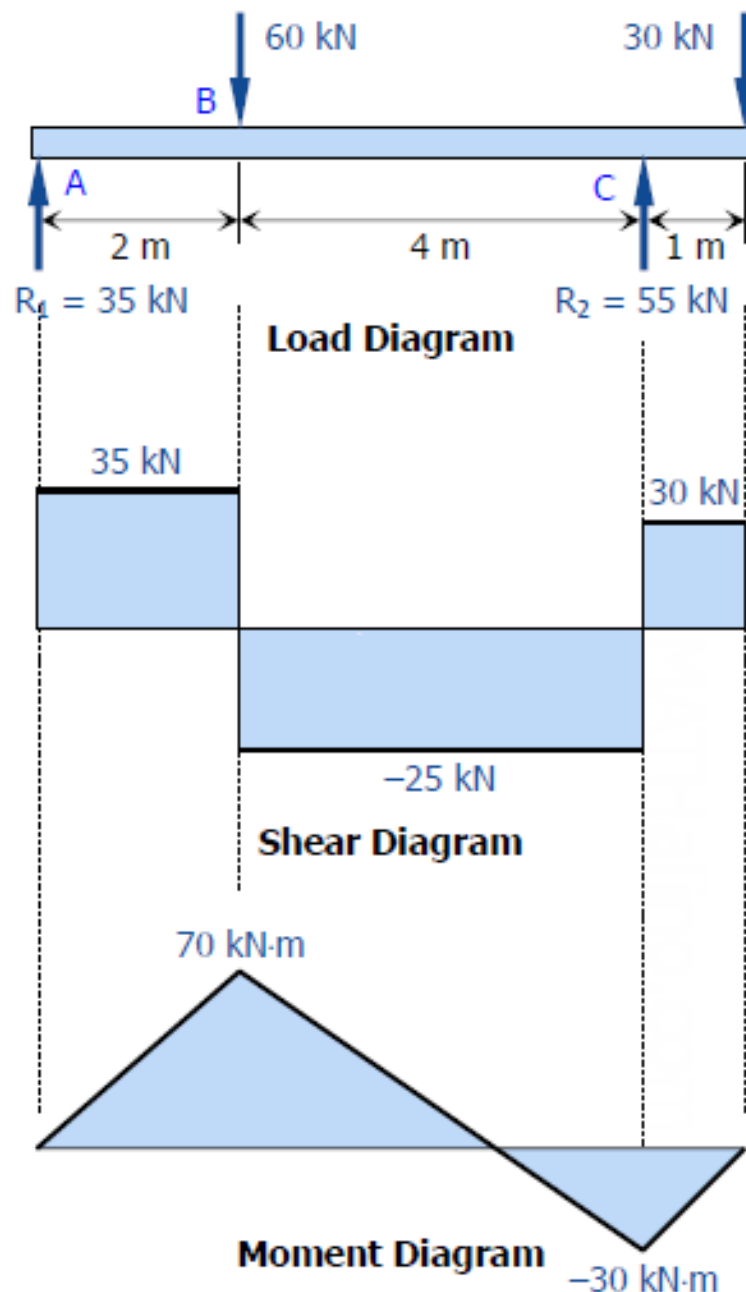
$$R_1 = 35 \text{ kN}$$

To draw the Shear Diagram:

1. $V_A = R_1 = 35 \text{ kN}$
2. $V_B = V_A + \text{Area in load diagram} - 60 \text{ kN}$
 $V_B = 35 + 0 - 60 = -25 \text{ kN}$
3. $V_C = V_B + \text{area in load diagram} + R_2$
 $V_C = -25 + 0 + 55 = 30 \text{ kN}$
4. $V_D = V_C + \text{Area in load diagram} - 30 \text{ kN}$
 $V_D = 30 + 0 - 30 = 0$

To draw the Moment Diagram:

1. $M_A = 0$
2. $M_B = M_A + \text{Area in shear diagram}$
 $M_B = 0 + 35(2) = 70 \text{ kN}\cdot\text{m}$
3. $M_C = M_B + \text{Area in shear diagram}$
 $M_C = 70 - 25(4) = -30 \text{ kN}\cdot\text{m}$
4. $M_D = M_C + \text{Area in shear diagram}$
 $M_D = -30 + 30(1) = 0$



Problem 426

Cantilever beam acted upon by a uniformly distributed load and a couple as shown in Fig. P-426.

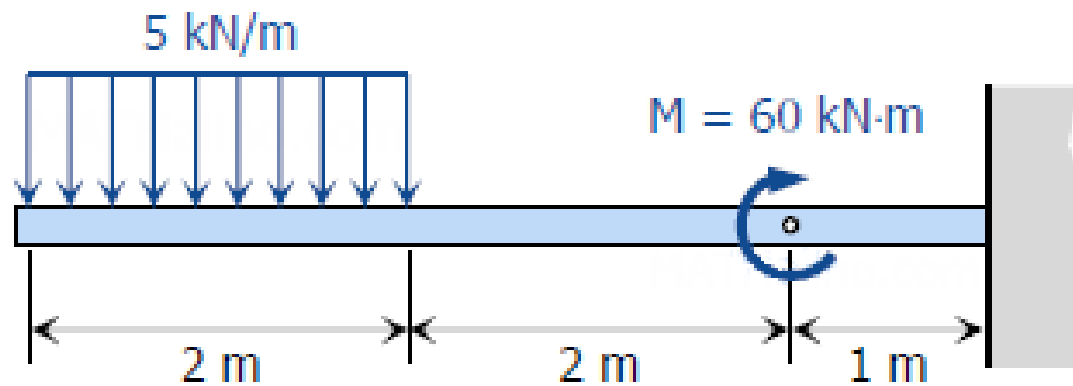


Figure P-426



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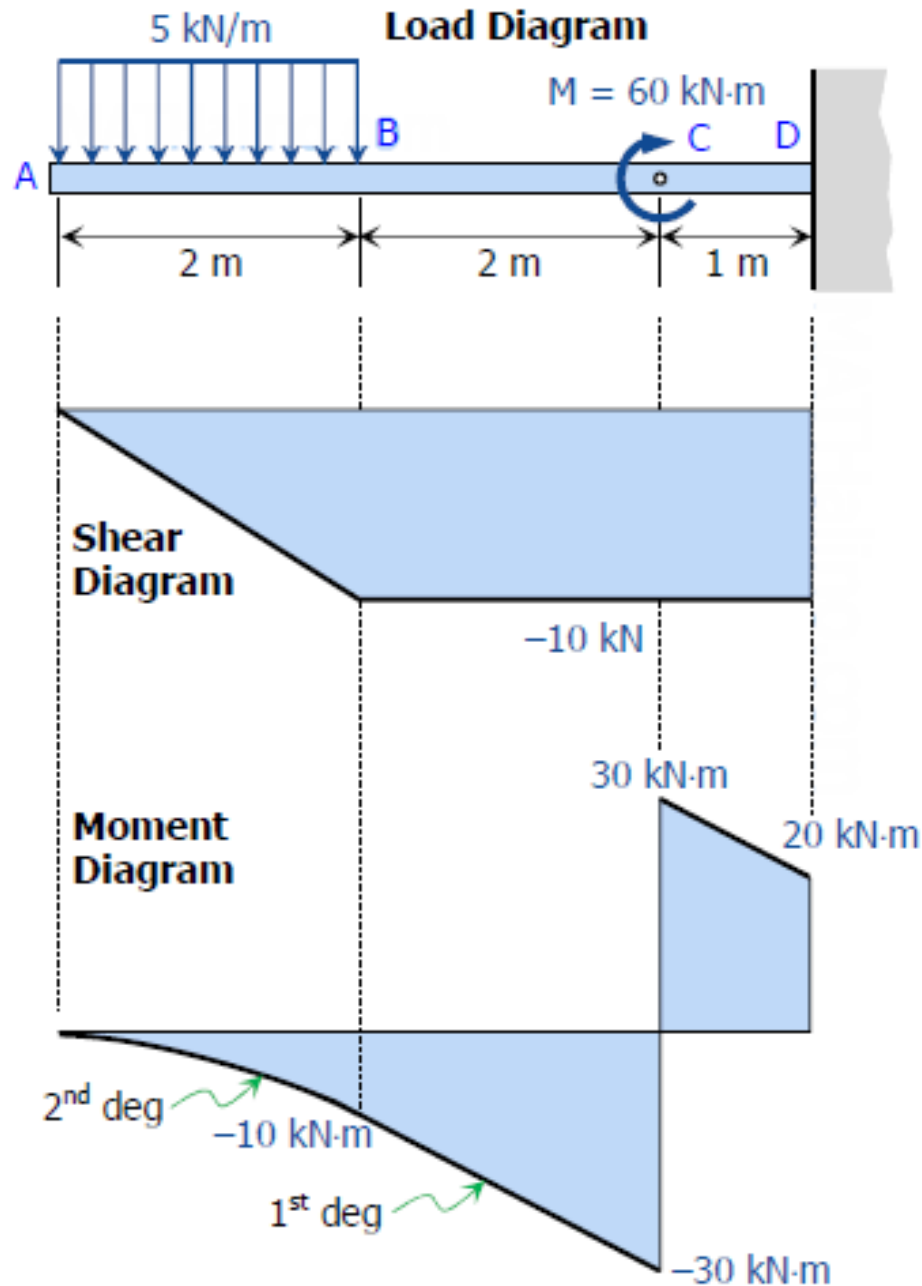
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To draw the Shear Diagram

1. $V_A = 0$
2. $V_B = V_A + \text{Area in load diagram}$
 $V_B = 0 - 5(2)$
 $V_B = -10 \text{ kN}$
3. $V_C = V_B + \text{Area in load diagram}$
 $V_C = -10 + 0$
 $V_C = -10 \text{ kN}$
4. $V_D = V_C + \text{Area in load diagram}$
 $V_D = -10 + 0$
 $V_D = -10 \text{ kN}$

To draw the Moment Diagram

1. $M_A = 0$
2. $M_B = M_A + \text{Area in shear diagram}$
 $M_B = 0 - \frac{1}{2}(2)(10)$
 $M_B = -10 \text{ kN}\cdot\text{m}$
3. $M_C = M_B + \text{Area in shear diagram}$
 $M_C = -10 - 10(2)$
 $M_C = -30 \text{ kN}\cdot\text{m}$
 $M_{C2} = -30 + M = -30 + 60 = 30 \text{ kN}\cdot\text{m}$
4. $M_D = M_{C2} + \text{Area in shear diagram}$
 $M_D = 30 - 10(1)$
 $M_D = 20 \text{ kN}\cdot\text{m}$



Problem 427

Beam loaded as shown in Fig. P-427.

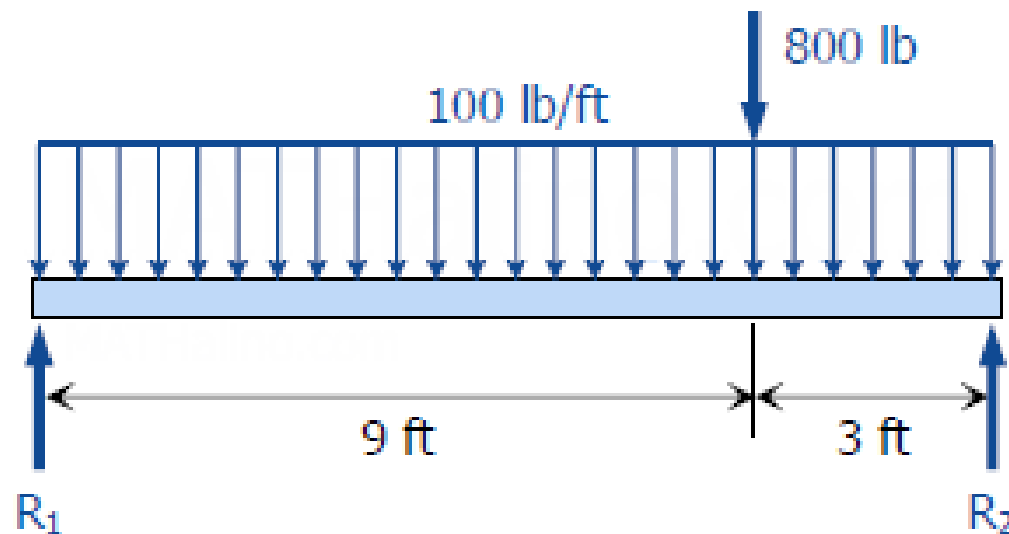


Figure P-427

$$\Sigma M_C = 0$$

$$12R_1 = 100(12)(6) + 800(3)$$

$$R_1 = 800 \text{ lb}$$

$$\Sigma M_A = 0$$

$$12R_2 = 100(12)(6) + 800(9)$$

$$R_2 = 1200 \text{ lb}$$



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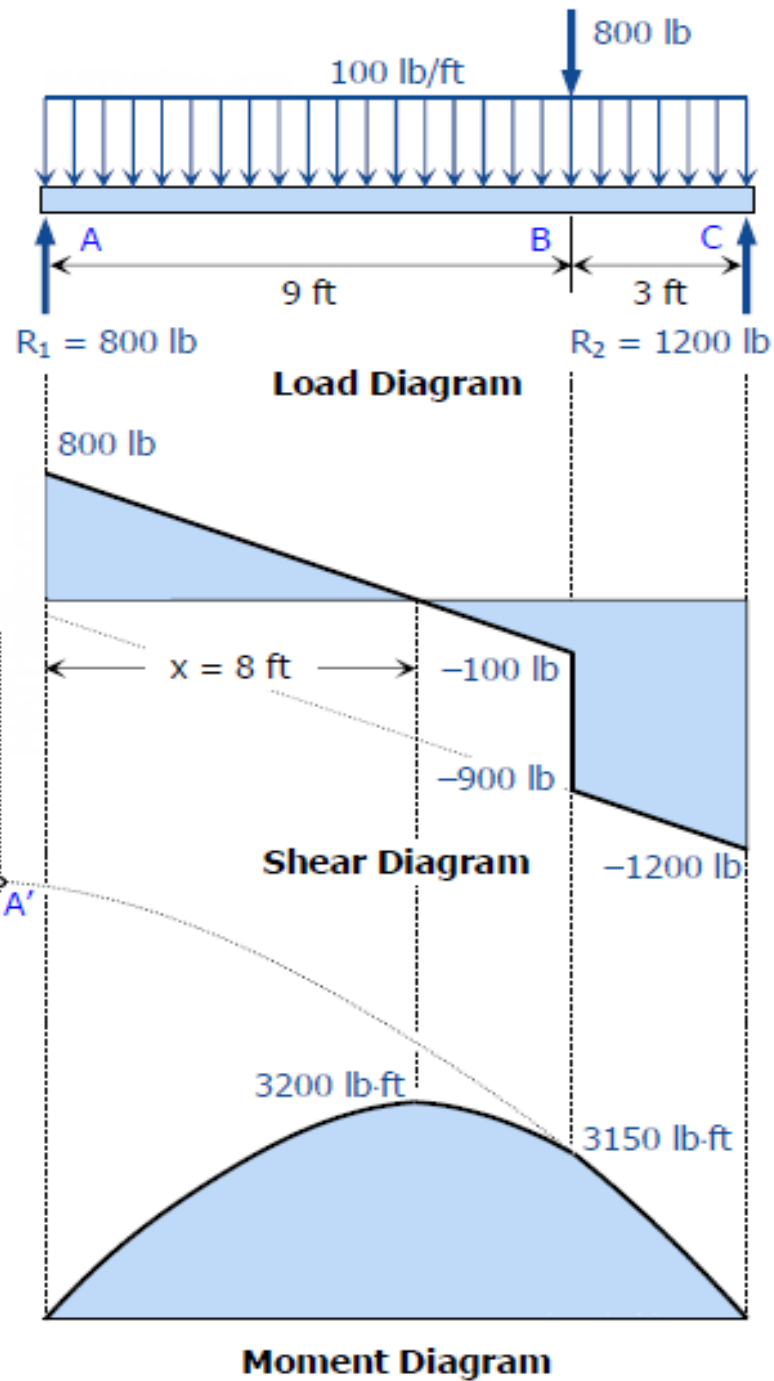
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To draw the Shear Diagram

1. $V_A = R_1 = 800 \text{ lb}$
2. $V_B = V_A + \text{Area in load diagram}$
 $V_B = 800 - 100(9)$
 $V_B = -100 \text{ lb}$
 $V_{B2} = -100 - 800 = -900 \text{ lb}$
3. $V_C = V_{B2} + \text{Area in load diagram}$
 $V_C = -900 - 100(3)$
 $V_C = -1200 \text{ lb}$
4. Solving for x:
 $x / 800 = (9 - x) / 100$
 $100x = 7200 - 800x$
 $x = 8 \text{ ft}$

To draw the Moment Diagram

1. $M_A = 0$
2. $M_x = M_A + \text{Area in shear diagram}$
 $M_x = 0 + \frac{1}{2}(8)(800) = 3200 \text{ lb}\cdot\text{ft};$
3. $M_B = M_x + \text{Area in shear diagram}$
 $M_B = 3200 - \frac{1}{2}(1)(100) = 3150 \text{ lb}\cdot\text{ft}$
4. $M_C = M_B + \text{Area in shear diagram}$
 $M_C = 3150 - \frac{1}{2}(900 + 1200)(3) = 0$
5. The moment curve BC is downward parabola with vertex at A'. A' is the location of zero shear for segment BC.





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